

WDW-5

Configuration-Space Differentiation and the Constraint-Native Wheeler–DeWitt Candidate Family

A source-faithful two-branch algebraic construction on the final WDW-4 interior

Amos Jay Maley

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Exact scope. The theorem constructs separately typed metric-scalar and tetrad-Standard-Model configuration-space candidate families, their regulated functional differentials, representations, and complete constraint actions on one algebraically invariant test domain. It reaches the regulator-indexed algebraic candidate level only. Regulator removal, anomaly freedom, a physical inner product, nonzero physical states, and relational observables remain WDW-6 burdens.

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Main result

On the exact compact regular branch accepted by integrated WDW-4, this paper constructs two separately typed configuration-space quantum-constraint candidate families: a metric-scalar/clock core and a torsion-free tetrad/spin/gauge/Higgs/chiral-fermion extension. Each family has a Sobolev configuration carrier, proper-redescription action, separating test algebra, independently defined first and second Fréchet differentials, CCR/CAR seed, smoothing family, ordering coordinate, and complete smeared constraint action on one algebraically invariant test domain. The Standard-Model branch uses a smooth-core strong dual with de-greewise completed exterior powers, so its configuration-dependent Dirac/Yukawa action and section-valued Hessian trace are total without a fixed-monomial restriction. Each family has a transpose action on its algebraic dual, formal joint-annihilator functor, and standing-preserving comparison predicate on an explicitly topologized ambient section carrier. The full fixed-rung seed is compared on its initial algebra, while its smooth-core representation is compared on the generated common domain. The result reaches proof-class level 2: a typed regulator-indexed algebraic operator family. It does not claim a Hilbert realization, regulator removal, quantum constraint closure, anomaly freedom, a physical inner product, a nonzero state, or separating physical observables.

Abstract

The Wheeler-DeWitt substitution

$$\pi^{ij}(x) \mapsto -i\hbar \frac{\delta}{\delta h_{ij}(x)}$$

does not by itself define a configuration carrier, a functional differential, a diagonal Hessian contraction, a representation, a common domain, or a quantum constraint system. This paper performs that missing candidate-level construction on the exact regular classical object exported by the accepted integrated WDW-4 theorem.

For $s > 7/2$ on the fixed compact boundaryless spin three-manifold Σ , we construct an open metric-scalar Sobolev chart and a distinct graded tetrad-Standard-Model chart. Proper spatial, Lorentz, and internal gauge redescrptions act on these charts and on separating cylinder/local algebras; descent is used only on the accepted regular free locus. The full six-direction $\mathfrak{spin}(1,3)$ Lorentz action is retained. First and second derivatives are Fréchet differentials with continuous Sobolev-dual types. A declared pivot-scale heat-smoothing family turns each branch's order-zero kinetic coefficient multiplier into a trace-class regulated Hessian contraction without any false $H^{-s} \rightarrow H^s$ regularity gain. The bosonic canonical pairs give algebraic CCR representations, while the Dirac-reduced chiral half-density bracket gives the graded CAR seed. That seed embeds into the strong dual of the smooth spinor core and its completed projective exterior powers, with completion only inside each finite particle degree.

The improved spatial, internal-Gauss, local-Lorentz, and branch Hamiltonian constraints are then defined for every allowed smearing, regulator, and ordering coordinate. Rungwise Dirac/Yukawa estimates, strong transposition, continuous $d\Gamma$ lifts, and a seminorm-convergent target-valued heat trace first prove every primitive total on a Bastiani-smooth finite-particle section algebra. Unguarded finite-primitive closure then produces a single domain Φ_5 preserved by every individual constraint action,

$$\widehat{C}_{\epsilon, \circ}[\alpha] : \Phi_5 \longrightarrow \Phi_5.$$

Its transpose is consequently defined on the full algebraic dual $\Phi_5^\times$, where the formal joint-annihilator functor is typed and may be the zero subspace. Each exterior degree carries the compact-open Bastiani topology and the finite-particle carrier its locally convex direct-sum topology. Candidate equivalence requires a bicontinuous matching package with explicit source, base, proper-groupoid, representation-algebra, primal/dual one-particle, and complete constraint-label maps. It preserves the full fixed-rung CCR/CAR seed on its initial algebra, the smooth-core representation on Φ_5 , every regulated constraint, fermionic rigging and degree filtration, boundary and charge status, clock/failure data, and the complete continuation ledger. Identity, inverse, and composition are componentwise, and the solution bijection follows from the bijective constraint square.

The construction occupies all fourteen theorem gates and all sixteen **QReg5** clauses. Hilbert-space density, invariant composition domains, adjoints, closability, regulator removal, quantum anomaly freedom, physical inner product, state nontriviality, and observables remain explicit WDW-6 burdens.

Keywords: Wheeler-DeWitt equation; canonical quantum gravity; configuration space; Sobolev manifold; functional differential; constraint quantization; Standard Model; AASC.

1. Exact question and proof class

1.1 Candidate closure, not quantum closure

The result answers one bounded question:

Given the exact regular classical constraint package exported by integrated WDW-4, can one construct a fully typed configuration-space differential and a complete regulator-indexed algebraic constraint family without borrowing a Hilbert measure, an anomaly theorem, or a physical state?

The answer is yes. The output is the target-indexed candidate tuple

$$\mathbf{Q}_{5b} = (\mathcal{U}_{5b}^s, \mathcal{G}_{5b}^{\text{prop}}, \Phi_{5b}, \Phi_{5b}^\times, \rho_{5b}, \mathcal{K}_b, \mathfrak{D}_b, \widehat{\mathbf{C}}_{5b}, \text{Sol}_{5b}, \sim_5, \mathfrak{G}_6), \quad b \in \{A, B\}.$$

The branch A object is the load-bearing metric-scalar/clock candidate. The branch B object retains the full torsion-free tetrad/spin/gauge/Higgs/chiral-fermion content. Their common packaging does not identify their configuration carriers or constraint actions.

The proof-class ladder is fixed:

1. typed functional differential expression;
2. regulator-indexed algebraic constraint family on a common test domain;
3. densely defined operators on a kinematical Hilbert space;
4. invariant, closable or closed common composition domain;
5. regulator removal or a controlled regulated endpoint;
6. anomaly-free joint quantum representation;
7. nontrivial physical state space and physical inner product;
8. separating physical or relational observables.

This paper proves levels 1 and 2. Levels 3–8 are not consequences of the notation used at level 2.

1.2 Dependency lock

The proof order is

nondegenerate determinate construction $\implies$ AASC kernel necessity
 $\implies$ fixed admissible physical interior $\implies$ accepted WDW-1 canonical projection
 $\implies$ accepted WDW-2 proper-redescription groupoid $\implies$ accepted WDW-3 refoliation closure
 $\implies$ accepted integrated WDW-4 classical occupancy $\implies$ WDW-5 candidate construction
 $\implies$ open WDW-6 survival tests.

AASC is not an optional interpretive overlay on independently complete physics. Determinacy, identity, comparison, admissible transformation, standing-bearing use, and irreversible fixation presuppose Admissibility, Standing, Reference, and Irreversibility. Every physical domain containing determinate carriers, states, records, comparisons, or endpoints therefore instantiates those kernel roles.

Kernel necessity governs the standing of the objects constructed below. It does not choose s , a configuration polarization, a smoothing profile, an ordering, a CAR polarization, a measure, or a rigging map. Those are positive mathematical burdens. Conversely, no downstream quantum candidate retro-authorizes the kernel or repairs an imported classical theorem.

2. Registered sources and exact classical input

2.1 Publication source lineage

The sole controlling predecessor input is the final integrated WDW-4 theorem. Three AASC manuscripts are registered below for provenance or downstream comparison; they are not used to construct or authorize a WDW-5 object:

Source	Licensed use
<i>Classical Matter Occupancy and Joint-Constraint Closure under AASC</i> (WDW-4)	Exact scalar/clock and torsion-free Standard Model classical input

Source	Licensed use
<i>Schrödinger Dynamics as the Unique Minimal First-Response Hamiltonian Projection of AASC Quantum Standing</i>	Downstream comparison only: projective-Hilbert and Hamiltonian role boundary; no theorem imported
<i>Admissibility and Constraint-Induced Rigidity of Physical Quotient Structure</i>	Quotient fidelity, local regular descent, and equivalence discipline
<i>Structural Unification of Gravity and Quantum Dynamics under AASC</i>	Canonical and Wheeler–DeWitt route classification and scope ledgers

The first comparison manuscript is downstream of the present ARC and supplies no premise here. W5.1 proves the required no-backflow statement locally; Sections 9–12 construct only the printed algebraic representation seed and common algebraic action domain; and Hilbert completion, density, closed graph domains, adjoints, and anomaly closure remain explicit WDW-6 burdens. The second manuscript supplies quotient fidelity, local regular descent, no-second-classifier discipline, and the prohibition on identifying formal similarity with physical equivalence. The third classifies canonical and Wheeler–DeWitt routes as richer-scope candidates that require explicit constraint, domain, record, source, and reportability ledgers. None of the three supplies a functor from the ADM phase space to a Hilbert space, a configuration-space Hessian, or the WDW-5 operators constructed here.

2.2 Complete integrated WDW-4 payload

Let z_0 be the accepted nonvacuum Einstein–Klein–Gordon datum, let $w_\delta = \mathcal{L}_\delta(\mathcal{I}_{4A \rightarrow 4B})$ be the accepted nonzero gauge-Higgs Standard-Model datum, and let

$$\mathcal{H}_{4B}^{\text{cov}}(w_\delta) = [\text{Dev}_{\text{EYMH}\phi}(w_\delta; \Psi_\delta = 0)]_{\cong_{t_\delta}}$$

be its source-attached pointed development class. The exact WDW-4B object is

$$\begin{aligned} \mathfrak{H}_{\text{W4B}}^{\text{cl}}(w_\delta) = & ((\mathcal{P}_{4B}^{\text{red}}, \{\cdot, \cdot\}_D), \\ & (\Phi_{4B}, \mathfrak{A}_{4B}^{\text{con}}), \\ & \mathcal{H}_{4B}^{\text{cov}}(w_\delta), \\ & (\mathcal{C}_{4B}^{\text{reg}}/\mathcal{R}_0, \text{singular guards}), \\ & \mathfrak{G}_{\text{target}}, \mathcal{A}_{\text{status}}, \\ & (\mathcal{U}_{\text{clk}}, \text{ClockFail}, \mathfrak{G}_{\text{quantum}})). \end{aligned}$$

The six displayed lines contain the accepted seven coordinates: the first two lines each form one coordinate, the fifth line contains the target and anomaly coordinates separately, and the final line is the clock/failure/quantum-guard coordinate.

The fixed target ledger is

$$\begin{aligned} \mathfrak{G}_{\text{target}} = & (\Sigma \text{ compact, connected, boundaryless, oriented, spin;} \\ & \text{second-order metric GR and oriented coframe } e; \omega = \omega_{\text{LC}}(e); \\ & \text{accepted scalar/clock subsystem with its failure boundary;} \\ & G_{\text{SM}} = (SU(3) \times SU(2) \times U(1))/\mathbb{Z}_6, P_0 \simeq \Sigma \times G_{\text{SM}}; \\ & N_{\text{gen}} = 3, \nu_R \text{ absent, } H \in (\mathbf{1}, \mathbf{2})_{1/2}; \\ & g_3, g_2, g_1, Y_u, Y_d, Y_e, V_H, \theta, \Lambda, \kappa \text{ fixed;} \\ & \text{proper spatial, Lorentz, and internal redescrptions;} \\ & \text{regular Sobolev branch, singular stabilizers retained;} \\ & \mathbb{Z}_2\text{-graded Dirac-Bergmann carrier;} \\ & \text{no boundary, nontrivial bundle, line insertion, or topology change}). \end{aligned}$$

The attached open-quantum ledger is

$\mathfrak{G}_{\text{quantum}} =$ (configuration differential and regulated constraints: WDW-5;
 ordering and candidate comparison: WDW-5;
 Hilbert completion and common closed domain: WDW-6;
 regulator removal and quantum constraint algebra: WDW-6;
 local/global/determinant/bordism anomalies: WDW-6;
 state nontriviality, inner product, rigging map: WDW-6;
 physical observables and quantum clock relation: WDW-6;
 boundary/inflow data: inactive under the fixed target guard).

With $\mathfrak{H}_{4A}^{\text{cl}}$ the accepted scalar package, the complete source is

$$\mathfrak{H}_4^{\text{cl}} = (\mathfrak{H}_{4A}^{\text{cl}}, \mathfrak{H}_{W4B}^{\text{cl}}(w_\delta), \mathcal{J}_{4A \rightarrow 4B}, \mathcal{L}_\delta).$$

The two classical zeroes remain different typed objects. In particular, the gravity-scalar projection of w_δ is not asserted to be z_0 or a separate zero of the pure WDW-4A constraint map.

2.3 Exact import and no-backflow

Theorem W5.1 (Exact classical import and no-backflow).

The tuple $\mathfrak{H}_4^{\text{cl}}$ defines a nonempty, source-attached, target-indexed classical input for WDW-5. Its two phase carriers, two constraint maps, pointed history, proper quotient, anomaly status, clock failure boundary, target ledger, and quantum continuation ledger are retained without identification or modification. Every construction in WDW-5 is a downstream functor of this registered input and cannot alter the truth value, hypotheses, or failure boundaries of WDW-1 through WDW-4.

Proof.

The final integrated publication source supplies z_0 , w_δ , the typed interface, and the lift, while expressly distinguishing their constraint maps and zero loci. The displayed reconstruction of $\mathfrak{H}_{W4B}^{\text{cl}}$ retains the pointed covariant-history class and all seven WDW-4B handoff coordinates. The two finite ledgers retain every fixed target and open quantum status.

Define the WDW-5 source projection

$$\text{src}_5(\mathbf{Q}_{5b}) = \mathfrak{H}_4^{\text{cl}} \quad (b = A, B).$$

Every later definition is parameterized by this source projection and by a candidate coordinate α ; no later map has codomain in the set of classical WDW-4 theorem statements. Hence there is no directed path from a WDW-5 candidate status back to an imported classical premise. The no-quantization-primacy and projection-backflow exclusions in the registered support corpus enforce the same direction semantically: a downstream projective or operator presentation cannot authorize the already fixed classical object or the kernel roles that make determinate use possible. Therefore the import is complete and no-backflow holds. $\square$

2.4 Frozen branch

Throughout,

$$\Sigma = \mathbb{H}^3/\Gamma_\Sigma, \quad \partial\Sigma = \emptyset, \quad s > \frac{7}{2}.$$

The value $s > 7/2$ gives $H^s \hookrightarrow C^2$ in dimension three and makes H^s a Banach algebra. Any use of a derived connection or curvature is typed in H^{s-1} or H^{s-2} , respectively. Spatial diffeomorphisms, gauge transformations, and local Lorentz transformations are recorded on the full scale $\{H^{s+k}\}_{k \geq 0}$. Local slices are used at a fixed Hilbert rung; infinitesimal diffeomorphism generators act on the smooth inverse-limit core, where their one-derivative loss is explicit.

Fixed-scope exclusions

No result below classifies noncompact, bounded, asymptotic, topology-changing, nontrivial-bundle, line-insertion, non-spin, torsionful, nonmetric, right-handed-neutrino, extra-Higgs, singular-quotient, or global-clock branches. These are target changes or separately gated continuations, not counterexamples within the fixed theorem domain.

3. Sobolev configuration carriers

3.1 Core metric-scalar chart

Fix the accepted reference datum $z_0 = (h_0, \pi_0; \phi_0, p_{\phi,0})$. Let

$$\mathcal{H}_g^s = H^s(\Sigma; S^2 T^* \Sigma), \quad \mathcal{H}_\phi^s = H^s(\Sigma; \mathbb{R}).$$

The metric cone

$$\text{Riem}^s(\Sigma) = \{h \in \mathcal{H}_g^s : h(x) > 0 \text{ for every } x\}$$

is open in $\mathcal{H}_g^s$. Indeed, $H^s \hookrightarrow C^0$, the smallest eigenvalue of h_0 has a positive minimum on compact Σ , and sufficiently small H^s perturbations remain pointwise positive.

Choose radii $r_g, r_\phi > 0$ small enough that

$$\mathcal{U}_{5A}^s = (B_{r_g}^{H^s}(h_0) \cap \text{Riem}^s(\Sigma)) \times B_{r_\phi}^{H^s}(\phi_0)$$

lies inside the accepted WDW-4A regular neighborhood and its clock domain where required. This is an open Hilbert-manifold chart modeled on $\mathcal{H}_g^s \oplus \mathcal{H}_\phi^s$. It is a prequotient chart: diffeomorphic points have not yet been identified.

The point z_0 includes momenta, whereas $\mathcal{U}_{5A}^s$ is the configuration projection. The source attachment is retained by the pointed pair

$$(\mathcal{U}_{5A}^s; q_{A,0} = (h_0, \phi_0); p_{A,0} = (\pi_0, p_{\phi,0})),$$

not by inserting the classical momenta into the quantum configuration coordinates.

For $k \geq 0$, intersection with the corresponding higher-regularity neighborhood defines $\mathcal{U}_{5A}^{s+k}$. The nested family has smooth inverse-limit core

$$\mathcal{U}_{5A}^\infty = \bigcap_{k \geq 0} \mathcal{U}_{5A}^{s+k}.$$

Fréchet differentials and heat estimates are proved rung by rung; no same-rung differentiability of Sobolev composition is assumed.

3.2 Tetrad-Standard-Model graded chart

Let e_δ be the accepted oriented nondegenerate spatial tetrad in w_δ . With $V \simeq \mathbb{R}^{1,3}$ the internal Minkowski space,

$$\mathcal{E}_+^s = \{e \in H^s(T^* \Sigma \otimes V) : h_{ij}(e) = \eta_{IJ} e_i^I e_j^J > 0, \text{ } e \text{ has the accepted orientation and time orientation}\}$$

is open by the same $H^s \hookrightarrow C^0$ argument: positivity gives rank three, while orientation and time orientation are open on the accepted component. The spin connection is not independent:

$$\omega(e) = \omega_{\text{LC}}(e) \in H^{s-1}.$$

On the fixed trivial G_{SM} -bundle P_0 , set

$$\begin{aligned} \mathcal{A}_{\text{SM}}^s &= \mathcal{A}_\delta + H^s(T^* \Sigma \otimes (\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1))), \\ \mathcal{H}_H^s &= H^s(\Sigma; E_H) \oplus H^s(\Sigma; \overline{E}_H), \quad \mathcal{H}_\phi^s = H^s(\Sigma; \mathbb{R}). \end{aligned}$$

The bosonic chart is a sufficiently small product neighborhood

$$\mathcal{U}_{5B,0}^s \subset \mathcal{E}_+^s \times \mathcal{A}_{\text{SM}}^s \times \mathcal{H}_H^s \times \mathcal{H}_\phi^s$$

centered at $(e_\delta, \mathcal{A}_\delta, H_\delta, H_\delta^\dagger, \phi_\delta)$ and contained in the accepted WDW-4B regular stratum.

The odd coordinate is not treated as an ordinary open set of commuting fields. Let

$$\mathcal{S}_{\text{SM}}^s = H^s(\Sigma; \mathbb{S}_s^{1/2} \otimes \mathcal{R}_{\text{SM}} \otimes |\Lambda|^{1/2})$$

be the direct sum of the three-generation chiral half-density bundles listed in WDW-4B. The local graded carrier is the ringed object

$$\mathcal{U}_{5B}^s = (\mathcal{U}_{5B,0}^s, C^\infty(\mathcal{U}_{5B,0}^s) \otimes_{\text{alg}} \Lambda_{\text{alg}}((\mathcal{S}_{\text{SM},\text{test}}^s)^*)).$$

Thus the even body is a Hilbert manifold and the odd directions are an algebraic exterior module. No analytic implicit-function theorem is invoked in an odd direction.

The higher-regularity even charts and odd test modules similarly define $\mathcal{U}_{5B}^{s+k}$ and the smooth inverse-limit core $\mathcal{U}_{5B}^\infty$. The derived spin connection lowers the rung by one and curvature lowers it by two.

3.3 Carrier theorem

Theorem W5.2 (Sobolev configuration-carrier theorem).

For $s > 7/2$, the accepted data z_0, w_δ have neighborhoods $\mathcal{U}_{5A}^s$ and $\mathcal{U}_{5B}^s$ with the following properties:

1. $\mathcal{U}_{5A}^s$ is an open metric-scalar Hilbert-manifold chart;
2. the even body of $\mathcal{U}_{5B}^s$ is an open coframe-connection-Higgs- clock Hilbert-manifold chart;
3. its odd Standard-Model half-density content is retained as a separately typed algebraic exterior direction;
4. the nested Sobolev scales have smooth inverse-limit cores;
5. $h(e) \in H^s$, $\omega_{\text{LC}}(e) \in H^{s-1}$, and every curvature or second-order classical coefficient used later has its declared H^{s-2} type;
6. orientation, spin structure, bundle, global form, matter inventory, topology, boundary status, and the local clock/failure boundary agree exactly with $\mathfrak{G}_{\text{target}}$.

Proof.

Sobolev embedding gives $H^s \hookrightarrow C^2$ and continuous pointwise multiplication. Positivity of $h(e) = e^*\eta$, rank three, and retention of the selected orientation/time-orientation component are open C^0 conditions; compactness turns the pointwise positive lower bound into a uniform one. The metric cone and oriented spatial-tetrad locus are therefore open in their H^s model spaces.

The gauge connection space is affine over an H^s Hilbert space, while the Higgs and clock spaces are linear Hilbert spaces. Their finite product is a Hilbert manifold. The maps $e \mapsto h(e)$ and $e \mapsto \omega_{\text{LC}}(e)$ are smooth on the nondegenerate locus by the H^s multiplication and inversion theorems; one spatial derivative is lost in ω , and one more in curvature. Since $s - 2 > 3/2$, the resulting curvature coefficients still multiply the fields that occur in the classical potentials.

The exterior algebra over the declared chiral test dual is a $\mathbb{Z}_2$ -graded algebraic extension of the even chart. It retains all fermion species even though the accepted classical witness has zero odd body. Intersecting the higher Sobolev rungs supplies the smooth core without changing the base Hilbert chart. Finally, choosing the neighborhoods inside the accepted regular strata imports no new topology, bundle, spin, global-form, boundary, or matter coordinate. The six claims follow. $\square$

4. Proper-redescription groupoids and local descent

4.1 Restricted actions

Let $\text{Diff}_0^{s+k+1}(\Sigma)$ denote the identity-component Sobolev diffeomorphisms retained at rung k by the accepted WDW-2 proper branch. On the core chart,

$$\varphi \cdot (h, \phi) = (\varphi^*h, \varphi^*\phi).$$

On the extension chart, the rung- k proper arrow set also includes

$$\mathcal{G}_{\text{int}}^{s+k+1} = H^{s+k+1}(\Sigma; G_{\text{SM}}), \quad \mathcal{G}_{\text{Lor}}^{s+k+1} = H^{s+k+1}(\Sigma; \text{Spin}^+(1, 3)),$$

restricted to the accepted global branch. For (φ, g, Λ) ,

$$\begin{aligned} e &\mapsto \Lambda \varphi^*e, & \mathcal{A} &\mapsto g \varphi^* \mathcal{A} g^{-1} - dg g^{-1}, \\ H &\mapsto \rho_H(g) \varphi^*H, & \Psi &\mapsto \rho_{\text{spin}}(\Lambda) \rho_{\text{SM}}(g) \varphi^*\Psi, \\ \phi &\mapsto \varphi^*\phi. \end{aligned}$$

All actions are restricted to arrows whose source and target lie in the chosen rung. The resulting compatible action groupoids at the base rung are $\mathcal{G}_{5A}^{\text{prop}} \rightrightarrows \mathcal{U}_{5A}^s$ and $\mathcal{G}_{5B}^{\text{prop}} \rightrightarrows \mathcal{U}_{5B}^s$.

Sobolev pullback is not treated as a differentiable same-rung map. At finite regularity the spatial fundamental expression has the typed loss

$$X_\xi : \mathcal{U}_{5b}^{s+k+1} \longrightarrow T\mathcal{U}_{5b}^{s+k}.$$

On the inverse-limit core, smooth ξ, λ, ϑ , with

$$\vartheta = \frac{1}{2} \vartheta^{IJ} J_{IJ} \in \Gamma^\infty(\mathfrak{spin}(1, 3)), \quad \vartheta^{IJ} = -\vartheta^{JI},$$

differentiate the actions and give

$$\begin{aligned} X_\xi q &= \mathcal{L}_\xi q, \\ X_\lambda \mathcal{A} &= -D_\lambda \mathcal{A}, \quad X_\lambda H = \rho_{H*}(\lambda)H, \quad X_\lambda \Psi = \rho_{\text{SM}*}(\lambda)\Psi, \\ X_\vartheta e &= \vartheta \cdot e, \quad X_\vartheta \Psi = \rho_{\text{spin}*}(\vartheta)\Psi. \end{aligned}$$

The antisymmetric pair IJ supplies all six Lorentz directions: J_{0a} are boosts and J_{ab} are rotations. Every displayed expression is a smooth tangent vector on the inverse-limit core; the finite-rung loss remains visible in the scale.

4.2 Free regular locus and guards

Let $\mathcal{U}_{5b}^{\text{fr}}$ be the intersection of the chart with the accepted free regular locus. The accepted slice theorem supplies local transversals S_q and local groupoid equivalences

$$\mathcal{G}_{5b}^{\text{prop}}|_{\mathcal{O}_q} \simeq (\mathcal{O}_q \cap S_q) \times \mathcal{G}_{5b,q}^{\text{prop}}.$$

On the free locus the isotropy is trivial and invariant functionals restrict to functionals on S_q . This is the only quotient chart used below.

At a point with a Killing field, residual gauge symmetry, or other stabilizer, no smooth manifold quotient is asserted. Improper transformations, boundary charges, large disconnected transformations, and arrows leaving the frozen chart are retained in the target and continuation ledgers rather than divided out.

4.3 Descent theorem

Theorem W5.3 (Proper-redescription groupoid descent theorem).

The accepted proper-redescription groupoids restrict to the two WDW-5 configuration scales, differentiate to the displayed fundamental fields on the smooth inverse-limit cores with one finite-rung spatial derivative loss, and act by graded algebra automorphisms on the test algebras constructed below. On $\mathcal{U}_{5b}^{\text{fr}}$, invariant functionals descend through the accepted local slice. No global smooth superspace, singular-stratum erasure, or quotient by an improper transformation is used.

Proof.

Pullback, the affine gauge action, and the finite-dimensional fiber representations preserve the declared Sobolev scale at the stated regularity. Composition and inversion follow from the underlying diffeomorphism, gauge, and spin groups, while restricting arrows to the charts produces a groupoid. The Lie derivative of a configuration coefficient consumes one spatial derivative, giving the displayed map from rung $s + k + 1$ to rung $s + k$. Passing to the inverse-limit core makes the Lie, covariant-gauge, and spin expressions smooth tangent fields without asserting false same-rung differentiability.

Precomposition of a smooth test functional with a groupoid arrow preserves finite probe dependence and the exterior degree, hence acts by graded algebra automorphisms. On the free regular locus, the imported local slice identifies each nearby orbit with one slice point. An invariant functional is constant on that orbit and therefore factors locally through the slice projection. No step applies this factorization at nontrivial isotropy or beyond the proper arrow set. $\square$

5. Tangent, cotangent, and canonical normalization

5.1 Sectorwise types

At $q_A = (h, \phi)$,

$$T_{q_A} \mathcal{U}_{5A}^s = H^s(S^2 T^* \Sigma) \oplus H^s(\Sigma),$$

and its continuous dual, using the density-valued distribution convention, is

$$T_{q_A}^* \mathcal{U}_{5A}^s = H^{-s}(S^2 T \Sigma \otimes \text{Dens}) \oplus H^{-s}(\text{Dens}).$$

The branch B even tangent is

$$\begin{aligned} T_{q_B} \mathcal{U}_{5B,0}^s &= H^s(T^* \Sigma \otimes \mathbb{R}^{1,3}) \oplus H^s(T^* \Sigma \otimes \mathfrak{g}_{\text{SM}}) \\ &\oplus H^s(E_H) \oplus H^s(\overline{E}_H) \oplus H^s(\Sigma), \end{aligned}$$

with the density-valued H^{-s} dual in the contragredient bundles. Odd derivatives take values in the graded algebraic dual of $\mathcal{S}_{\text{SM},\text{test}}^s$.

For p of the dual density type and v of the tangent type,

$$\langle p, v \rangle_{\Sigma} = \int_{\Sigma} p \cdot v$$

denotes the distributional pairing. A reference density is used only to write Sobolev norms; the displayed integral pairs an intrinsic density with a scalar contraction.

5.2 Imported symplectic coefficients

On branch A , the accepted weak canonical form is

$$\Omega_{5A} = \int_{\Sigma} (\delta\pi^{ij} \wedge \delta h_{ij} + \delta p_{\phi} \wedge \delta\phi).$$

The gravitational and scalar Hamiltonian densities are

$$\begin{aligned}\mathcal{H}_{\perp}^g &= \frac{2\kappa}{\sqrt{h}} \left(\pi_{ij} \pi^{ij} - \frac{1}{2} \pi^2 \right) - \frac{\sqrt{h}}{2\kappa} (R(h) - 2\Lambda), \\ \mathcal{H}_{\perp}^{\phi} &= \frac{p_{\phi}^2}{2\sqrt{h}} + \frac{\sqrt{h}}{2} (D_i \phi D^i \phi + m^2 \phi^2).\end{aligned}$$

Thus the inverse DeWitt tensor in the accepted convention is

$$\mathcal{G}_{ijkl}(h) = \frac{1}{2\sqrt{h}} (h_{ik} h_{jl} + h_{il} h_{jk} - h_{ij} h_{kl}),$$

and the gravitational kinetic term is $2\kappa \mathcal{G}_{ijkl} \pi^{ij} \pi^{kl}$.

On branch B , the even canonical one-form is

$$\begin{aligned}\Theta_{5B,0} &= \int_{\Sigma} (P^i{}_I \delta e_i{}^I + p_{\phi} \delta\phi + \sum_{r=3,2,Y} E_r{}^i \cdot \delta \mathcal{A}_{r,i} \\ &\quad + \Pi_H \delta H + \delta H^{\dagger} \Pi_{H^{\dagger}}),\end{aligned}$$

and the reduced odd bracket is

$$\{\Psi_{\mathcal{A}}(x), \Psi_{\mathcal{B}}^{\dagger}(y)\}_D = -i \delta_{\mathcal{AB}} \delta_{\Sigma}(x, y).$$

The independent-complex-coordinate convention is used for $H, H^{\dagger}$ during differentiation and restricted to the real section afterward.

5.3 Pairing theorem

Theorem W5.4 (Configuration tangent/cotangent and symplectic pairing theorem).

The two WDW-5 carriers have continuous sectorwise tangent and cotangent types with intrinsic density pairings. The accepted WDW-1/4 symplectic forms fix the bosonic momentum rule

$$\hat{p}(v) = -i\hbar d_v$$

and the WDW-4B Dirac bracket fixes the odd CAR normalization

$$\{\widehat{\Psi}(u), \widehat{\Psi}^{\dagger}(v)\}_+ = \hbar \langle v, u \rangle.$$

No continuous dual is identified with a pointwise field without the displayed density pairing.

Proof.

Each chart is open in a finite product of Sobolev Hilbert spaces, so its tangent is the model space. Sobolev duality identifies the continuous dual with the corresponding H^{-s} distributional density bundle. The intrinsic pairing is nondegenerate because test fields are dense and separates nonzero distributions.

For a canonical pair with $\Omega = \int \delta p \wedge \delta q$, the Hamiltonian convention $\iota_{X_H} \Omega = -dH$ gives $\{q(\lambda), p(v)\} = \lambda(v)$.

The algebraic quantization rule $[\hat{q}, \hat{p}] = i\hbar \widehat{\{q, p\}}$ is therefore realized by configuration multiplication and $-i\hbar d_v$. For the reduced odd bracket, multiplication by $i\hbar$ gives $i\hbar(-i)\langle v, u \rangle = \hbar \langle v, u \rangle$, fixing the CAR coefficient. These calculations use only the declared pairings and do not invoke a Riesz map or functional measure. $\square$

6. Separating test algebras and variational calculus

6.1 Probe systems

Every Sobolev model space used above is separable. For each even sector $\mathcal{H}_r^s$, choose a countable total family $\{u_{r,n}\}_{n \geq 1} \subset \mathcal{H}_r^s$ and define

$$\ell_{r,n}(v) = \langle u_{r,n}, v \rangle_{\mathcal{H}_r^s}.$$

If every $\ell_{r,n}(v)$ vanishes, density of the span of the $u_{r,n}$ implies $v = 0$. The union over the finitely many sectors therefore separates the tangent model space and, by affine translation, each selected chart. The probes restrict to the smooth inverse-limit core and remain separating.

At a chart center q_* , an even cylinder functional is

$$F(q) = f(\ell_{\alpha_1}(q - q_*), \dots, \ell_{\alpha_n}(q - q_*)), \quad f \in C^\infty(\mathbb{R}^n; \mathbb{C}).$$

When localization is needed, F is multiplied by a smooth function of finitely many probes whose support is contained in a finite-dimensional coordinate box lying inside the chart. This is cylinder-local support. It does not assert the existence of a nonzero compactly supported function on an infinite-dimensional Hilbert chart.

Let Cyl_b^∞ be the unital algebra generated by these functionals, restricted to $\mathcal{U}_{5b}^\infty$ for the constraint action. Their Fréchet derivatives remain controlled at each Hilbert rung. Let $\mathcal{L}_b$ be the algebra generated by the inherited smeared local potentials whose first two variations are differential expressions of finite order on the chosen Sobolev chart. The initial even test algebra is

$$\mathcal{A}_b^{(0)} = \text{Cyl}_b^\infty \cdot \mathcal{L}_b.$$

For branch B ,

$$\mathcal{A}_{5B}^{(0)} = \mathcal{A}_{B,0}^{(0)} \otimes_{\text{alg}} \Lambda_{\text{alg}}((\mathcal{S}_{\text{SM,test}}^s)^*).$$

Every element has finite exterior degree, although the full algebra contains all finite degrees.

6.2 First and second Fréchet differentials

For a cylinder functional as above and $v, w \in T_q \mathcal{U}^s$,

$$dF(q)[v] = \sum_{a=1}^n \partial_a f(\ell(q - q_*)) \ell_{\alpha_a}(v),$$

$$d^2 F(q)[v, w] = \sum_{a,c=1}^n \partial_{ac} f(\ell(q - q_*)) \ell_{\alpha_a}(v) \ell_{\alpha_c}(w).$$

On a bounded finite-probe coordinate set,

$$|dF(q)[v]| \leq \left(\sum_a |\partial_a f| \|\ell_{\alpha_a}\| \right) \|v\|_{H^s},$$

and

$$|d^2 F(q)[v, w]| \leq \left(\sum_{a,c} |\partial_{ac} f| \|\ell_{\alpha_a}\| \|\ell_{\alpha_c}\| \right) \|v\|_{H^s} \|w\|_{H^s}.$$

The derivative maps are consequently continuous. Products with $\mathcal{L}_b$ obey the ordinary Fréchet Leibniz rules; the selected regularity makes their coefficient multiplications and finite-order variations continuous between the recorded Sobolev spaces.

For an odd generator η and a monomial $\eta_1 \wedge \dots \wedge \eta_k$, define the left contraction

$$\partial_\eta^L(\eta_1 \wedge \dots \wedge \eta_k) = \sum_{j=1}^k (-1)^{j-1} \langle \eta, \eta_j \rangle \eta_1 \wedge \dots \widehat{\eta_j} \dots \wedge \eta_k.$$

The right derivative is

$$\partial_\eta^R M = (-1)^{|M|-1} \partial_\eta^L M$$

on a homogeneous monomial M . These algebraic maps have the graded Leibniz signs and lower exterior degree by one. Their normalization agrees with the right-differentiation convention in the accepted WDW-4B second-class reduction.

6.3 Differential theorem

Theorem W5.5 (Cylinder/local differential theorem).

The core and Standard-Model charts admit test algebras that:

1. separate every declared even configuration direction;
2. retain every finite odd chiral direction;
3. have continuous first and second Fréchet differentials with the displayed tangent-dual and continuous-bilinear codomains;
4. support the declared left and right graded derivatives; and
5. use cylinder-local rather than fictitious infinite-dimensional compact support.

Proof.

Separability supplies the total probe families. If two chart points differ, their affine difference is nonzero in at least one sector and is detected by one total probe. Hence the cylinder algebra separates points. The exterior generators distinguish the declared odd test directions algebraically.

The chain rule gives the two displayed derivative formulas. Their norm estimates prove continuity, and the product rule extends the result to $\mathcal{A}_b^{(0)}$ because the local generators have finite-order continuous variations at the frozen Sobolev threshold. The contraction formula gives well-defined left derivatives on every finite exterior monomial; the right formula supplies the required parity conversion. Finite-probe localization never appeals to local compactness of the full Hilbert chart. $\square$

7. The finite-rank boundary

The accepted covariant-differentiation theorem concerns a finite-rank bundle $E \rightarrow M$ and a spacetime connection

$$\nabla : \Gamma(E) \longrightarrow \Gamma(T^*M \otimes E).$$

Its input is a section over one spacetime point and its output carries one spacetime cotangent index. In contrast, for

$$F : \mathcal{U}_{5b}^s \longrightarrow \mathbb{C},$$

the derivative is

$$dF(q) \in (T_q \mathcal{U}_{5b}^s)^*,$$

a distributional functional on an infinite-dimensional configuration tangent. The bases, fibers, codomains, topologies, and differentiation directions are different.

The finite-rank theorem still supplies valid downstream spacetime coefficients: $D_i \phi$, $\omega_{LC}(e)$, gauge-covariant derivatives, curvatures, and bundle transport. It supplies no map

$$\nabla^{LC} \mapsto \frac{\delta}{\delta h}$$

and no identity-principal-symbol theorem on configuration space.

7.1 Independent-differential theorem

Theorem W5.6 (Finite-rank nonextension and independent differential theorem).

Every configuration differential used in WDW-5 is obtained from the Sobolev-chart smooth structure and the probe/local algebra of Theorem W5.5. The finite-rank spacetime connection enters only inside the inherited local coefficients and bundle operators. No configuration derivative, Hessian, functional measure, field-space connection, or diagonal contraction follows from the finite-rank theorem.

Proof.

A finite-rank covariant derivative has domain $\Gamma(E)$ and codomain $\Gamma(T^*M \otimes E)$; a configuration derivative has domain a function on an open subset of a Sobolev space and codomain its continuous dual. There is no canonical map between these types. The first and second WDW-5 differentials have already been constructed by the Banach-space chain rule in Theorem W5.5, without invoking a spacetime connection.

When D_i , F_{ij} , $R(h)$, or $\omega_{LC}(e)$ occurs in a local coefficient, its finite-rank derivative is evaluated before that coefficient is multiplied into a configuration functional. Varying the coefficient with respect to q is then a separate Fréchet derivative of the coefficient map. This order establishes the claimed boundary and excludes an untyped extension. $\square$

8. Smoothing kernels and regulated Hessian contraction

8.1 Reference elliptic family

Choose the accepted smooth reference metric h_* in the chart, the fixed reference bundle metrics, and their compatible reference connections. For every even configuration sector E_r , let

$$L_{r,*} = 1 + \nabla_{r,*}^* \nabla_{r,*}$$

on $L^2(\Sigma, E_r; d\mu_*)$. It is positive, self-adjoint, elliptic, and has discrete spectrum on compact Σ . Set

$$S_{\epsilon,r} = e^{-\epsilon^2 L_{r,*}/2}, \quad K_{\epsilon,r} = S_{\epsilon,r} S_{\epsilon,r}^* = e^{-\epsilon^2 L_{r,*}}, \quad \epsilon > 0.$$

The heat kernel

$$K_{\epsilon,r}(x, y) : (E_r)_y \otimes |\Lambda|_y \longrightarrow (E_r)_x$$

includes the bundle-index transport and reference-density convention. Adjoints are taken in the declared $L^2(d\mu_*)$ structures. Heat smoothing maps H^{-m} continuously into H^k for all finite m, k and is trace class on L^2 for every $\epsilon > 0$.

The collection

$$\mathcal{K}_b = (h_*, \{\nabla_{r,*}, L_{r,*}, S_{\epsilon,r}, K_{\epsilon,r}\}_r)$$

is a candidate coordinate. A different reference, profile, or bundle transport is not declared equivalent by notation.

8.2 Kinetic tensors

Let $\mathcal{E}_{b,\bar{0}}$ be the direct sum of the even configuration bundles on branch b , written with the reference density and bundle metrics fixed in $\mathcal{K}_b$. For every $r \geq s$, use the pivot triple

$$H^r(\mathcal{E}_{b,\bar{0}}) \hookrightarrow \mathcal{L}_b^2 \hookrightarrow H^{-r}(\mathcal{E}_{b,\bar{0}}).$$

The density convention identifies the distributional cotangent with the corresponding H^{-r} coefficient bundle; it does not identify H^{-r} with H^r .

The branch A pointwise kinetic form on smooth momenta is

$$\mathbb{K}_A(q_A)[(p, p_\phi), (p, p_\phi)] = \frac{2\kappa}{\sqrt{h}} \left(p_{ij} p^{ij} - \frac{1}{2} p^2 \right) + \frac{p_\phi^2}{2\sqrt{h}}.$$

After the reference-density conversion and finite-fiber index change, its polarization defines, for smooth-core q and $N \in C^\infty(\Sigma)$, the order-zero multiplier

$$M_{N\mathbb{K}_A(q)} : H^{-r}(\mathcal{E}_{A,\bar{0}}^*) \longrightarrow H^{-r}(\mathcal{E}_{A,\bar{0}}) \quad (r \geq s).$$

This multiplier changes tensor type but gains no Sobolev derivatives. In particular, no map $H^{-s} \rightarrow H^s$, and no quadratic evaluation on an arbitrary H^{-s} distribution, is asserted. Indefiniteness of the DeWitt block is retained; positivity is neither needed nor asserted.

For branch B , let $\mathbb{K}_B(q_B)$ be the polarization of the exact even quadratic momentum part of the WDW-4B Hamiltonian on smooth momenta:

$$\mathcal{T}_B = \mathcal{T}_g(e, P) + \frac{p_\phi^2}{2\sqrt{h}} + \sum_{r=3,2,Y} \frac{E_r^i \cdot E_{r,i}}{2\sqrt{h}} + \frac{\Pi_H \Pi_{H^\dagger}}{\sqrt{h}}.$$

Here $\mathcal{T}_g(e, P)$ is the coframe pullback of the accepted metric kinetic term on the Lorentz-reduced sector; its Lorentz-null directions are handled by the separate Lorentz constraint, not inverted. The $(H, H^\dagger)$ block is the symmetric cross-polarization of the displayed product. Fermions are first order and do not enter $\mathbb{K}_B$. Exactly as on branch A , multiplication by $N\mathbb{K}_B(q)$ is an order-zero bounded map $H^{-r} \rightarrow H^{-r}$ between the correctly indexed distributional bundles. It is not a cotangent-to- H^r regularizer.

8.3 Operator trace

For $F \in \mathcal{A}_b^{(0)}$, or more generally for a Bastiani-smooth functional on the smooth inverse-limit core, continuity of $d^2 F(q)$ supplies a finite rung $r \geq s$, locally in q , and a bounded Hessian representative

$$B_{F,r}(q) : H^r(\mathcal{E}_{b,\bar{0}}) \longrightarrow H^{-r}(\mathcal{E}_{b,\bar{0}}^*).$$

The representative is the unique continuous extension from smooth tangent directions, which are dense in the selected H^r rung. For the initial Hilbert-chart algebra one may take $r = s$. Let

$$S_{\epsilon,b}^{(r)} : \mathcal{L}_b^2 \longrightarrow H^r(\mathcal{E}_{b,\bar{0}}), \quad S_{\epsilon,b}^{\vee,(r)} : H^{-r}(\mathcal{E}_{b,\bar{0}}) \longrightarrow \mathcal{L}_b^2$$

be the heat map and its pivot transpose. Both are Hilbert–Schmidt for $\epsilon > 0$. With N a lapse smearing, define

$$\text{Contr}_{\mathbb{K}_b, K_\epsilon} d^2 F(q)[N] \stackrel{\text{def}}{=} \text{Tr}_{\mathcal{L}_b^2} \left(S_{\epsilon,b}^{\vee,(r)} M_{N\mathbb{K}_b(q)} B_{F,r}(q) S_{\epsilon,b}^{(r)} \right),$$

where $\mathcal{L}_b^2$ is the direct sum of the reference L^2 sector spaces. Thus every arrow in the decisive composition is

$$\mathcal{L}_b^2 \xrightarrow{S_{\epsilon,b}^{(r)}} H^r \xrightarrow{B_{F,r}(q)} H^{-r} \xrightarrow{M_{N\mathbb{K}_b(q)}} H^{-r} \xrightarrow{S_{\epsilon,b}^{\vee,(r)}} \mathcal{L}_b^2.$$

The two outer arrows are Hilbert–Schmidt and the two middle arrows are bounded. The ideal property therefore makes the composition trace class. If two finite rungs control the same Hessian, compatibility of the Sobolev inclusions and heat operator shows that the trace is unchanged.

For a pure cylinder functional this is explicitly

$$\sum_{a,c} \partial_{ac} f \left\langle S_{\epsilon,b}^{\vee,(s)} \ell_{\alpha_c}, S_{\epsilon,b}^{\vee,(s)} M_{N\mathbb{K}_b(q)} \ell_{\alpha_a} \right\rangle.$$

For a local generator, its Hessian is a finite-order distributional differential operator and the two heat factors again make the composition trace class after choosing a rung above its finite order. The same trace-ideal estimate applied after every parameter derivative proves smooth dependence on q : the k -th derivative contains only finitely many terms built from $d^{k+2}F$, derivatives of the coefficient multiplier, and the fixed heat maps. The formula is a regulated contraction for $\epsilon > 0$; no value at $\epsilon = 0$ is assigned.

The same construction has a degreewise vector-valued form needed for the fermionic section carrier below. Let V be a complete locally convex space, let p be a continuous seminorm on V , and let

$$F : \mathcal{U}_{5b}^\infty \longrightarrow V$$

be Bastiani smooth. Suppose that, locally in q , a finite rung r controls its second derivative:

$$p(d^2 F(q)[v, w]) \leq C_{p,q} \|v\|_{H^r} \|w\|_{H^r}.$$

Let $T_{\epsilon,b}^{(r)} : \mathcal{L}_b^2 \rightarrow H^r$ denote the second Hilbert–Schmidt tangent leg obtained by moving $S_{\epsilon,b}^{\vee,(r)} M_{N\mathbb{K}_b(q)}$ across the declared pivot pairing. The ideal property gives

$$\|T_{\epsilon,b}^{(r)}\|_2 \leq \|M_{N\mathbb{K}_b(q)}\| \|S_{\epsilon,b}^{\vee,(r)}\|_2.$$

For an orthonormal basis (e_k) of $\mathcal{L}_b^2$, define

$$\text{Contr}_{\mathbb{K}_b, K_\epsilon}^V d^2 F(q)[N] \stackrel{\text{def}}{=} \sum_k d^2 F(q) \left[S_{\epsilon,b}^{(r)} e_k, T_{\epsilon,b}^{(r)} e_k \right].$$

The series is independent of the chosen orthonormal basis and satisfies

$$\sum_k p \left(d^2 F(q) \left[S_{\epsilon,b}^{(r)} e_k, T_{\epsilon,b}^{(r)} e_k \right] \right) \leq C_{p,q} \|S_{\epsilon,b}^{(r)}\|_2 \|T_{\epsilon,b}^{(r)}\|_2.$$

It therefore converges in V . The same estimate after each parameter derivative, locally uniformly in q , proves that the resulting V -valued section is Bastiani smooth. For $V = \mathbb{C}$ this is the operator trace already displayed.

8.4 Regulator theorem

Theorem W5.7 (Regulator and field-space contraction theorem).

For every branch, allowed lapse N , candidate coordinate $\mathcal{K}_b$, and $\epsilon > 0$, the displayed trace defines a finite smooth complex functional on the initial cylinder/local algebra. More generally, it defines a smooth target-valued contraction for every Bastiani-smooth section whose Hessian is locally controlled at a finite rung and whose target is complete. In particular, it acts degreewise on the completed exterior-section carrier constructed in Section 12. It retains all tensor, density, bundle, DeWitt-signature, gauge, Higgs-cross, exterior degree, and branch types. Reference-background, smoothing-profile, transport, pivot, and adjoint data remain explicit candidate coordinates.

Proof.

Elliptic heat calculus on compact Σ makes every $S_{\epsilon,r}$ smoothing; as a map from L^2 to any fixed positive Sobolev rung it is Hilbert–Schmidt, as is its pivot transpose back from the negative rung. The Hessian of a Hilbert-chart test functional is bounded $H^s \rightarrow H^{-s}$. On the inverse-limit core, continuity of each Bastiani derivative supplies a locally controlling finite rung. The Hessian of each admitted local generator has finite differential order and coefficients in a Sobolev algebra above the multiplication threshold.

The classical kinetic coefficient is only the bounded order-zero $H^{-r} \rightarrow H^{-r}$ multiplier displayed above. Hence

$$\|S_{\epsilon,b}^{\vee,(r)} M_{N\mathbb{K}_b(q)} B_{F,r}(q) S_{\epsilon,b}^{(r)}\|_1 \leq \|S_{\epsilon,b}^{\vee,(r)}\|_2 \|M_{N\mathbb{K}_b(q)}\| \|B_{F,r}(q)\| \|S_{\epsilon,b}^{(r)}\|_2.$$

The right side is finite. Applying the same estimate to every derivative of the product proves smoothness of the trace functional.

For a complete locally convex target, continuity of $d^2F(q)$ gives the displayed seminorm bound. Cauchy–Schwarz for the two Hilbert–Schmidt legs then gives absolute convergence in every continuous target seminorm. Completeness supplies the sum. Polarization and the scalar trace identity give basis independence. Differentiating the series introduces only finitely many Leibniz terms at each order; the same locally uniform estimate permits termwise differentiation and proves Bastiani smoothness.

The bundle heat kernels carry the source-to-target fiber maps, while the reference density and adjoint are fixed in $\mathcal{K}_b$. The kinetic coefficient multipliers are exactly the polarizations of the inherited classical momentum quadratics; no positive-definite replacement of the DeWitt block and no false Sobolev regularity gain occurs. Since the construction is defined only for $\epsilon > 0$, it makes no unregulated diagonal claim. $\square$

9. Algebraic CCR and CAR representation seeds

9.1 Bosonic configuration polarization

Let $\mathcal{H}_b^s$ denote the direct sum of the even branch- b configuration model spaces and let $(\mathcal{H}_b^s)^*$ be its continuous density dual. For $\lambda \in (\mathcal{H}_b^s)^*$ and $v \in \mathcal{H}_b^s$, define on $\mathcal{A}_b^{(0)}$

$$(\widehat{Q}_b[\lambda]F)(q) = \lambda(q - q_*)F(q), \quad (\widehat{P}_b[v]F)(q) = -i\hbar dF(q)[v].$$

The chart-center subtraction changes no commutator. Direct differentiation gives

$$\begin{aligned} [\widehat{Q}_b[\lambda], \widehat{Q}_b[\mu]] &= 0, \\ [\widehat{P}_b[v], \widehat{P}_b[w]] &= 0, \\ [\widehat{Q}_b[\lambda], \widehat{P}_b[v]] &= i\hbar \lambda(v) \mathbf{1}. \end{aligned}$$

These are algebraic relations on the test algebra. They do not require a measure, an adjoint, or a Stone-von Neumann uniqueness theorem.

Sectorwise, branch A contains the metric and scalar pairs

$$(\widehat{h}[\lambda_g], \widehat{\pi}[v_g]), \quad (\widehat{\phi}[\lambda_\phi], \widehat{p}_\phi[v_\phi]).$$

Branch B contains the coframe, scalar, three gauge, and independent Higgs-coordinate pairs

$$\begin{aligned} (\widehat{e}, \widehat{P}), \quad (\widehat{\phi}, \widehat{p}_\phi), \quad (\widehat{\mathcal{A}}_r, \widehat{E}_r) \quad (r = 3, 2, Y), \\ (\widehat{H}, \widehat{\Pi}_H), \quad (\widehat{H}^\dagger, \widehat{\Pi}_{H^\dagger}). \end{aligned}$$

Each coefficient and density pairing is the one displayed in the accepted canonical one-form.

9.2 Graded holomorphic seed

Let

$$\mathfrak{f} = (\mathcal{S}_{\text{SM}, \text{test}}^s)^*, \quad \Phi_{\text{odd}} = \Lambda_{\text{alg}}(\mathfrak{f}).$$

For $\eta \in \mathfrak{f}$ and $u \in \mathcal{S}_{\text{SM}, \text{test}}^s$, set

$$\widehat{\Psi}[\eta]M = \eta \wedge M, \quad \widehat{\Psi}^\dagger[u]M = \hbar \iota_u M.$$

Exterior multiplication and contraction obey

$$\begin{aligned} \{\widehat{\Psi}[\eta], \widehat{\Psi}[\zeta]\}_+ &= 0, \quad \{\widehat{\Psi}^\dagger[u], \widehat{\Psi}^\dagger[v]\}_+ = 0, \\ \{\widehat{\Psi}[\eta], \widehat{\Psi}^\dagger[u]\}_+ &= \hbar \eta(u) \mathbf{1}. \end{aligned}$$

This is precisely $i\hbar$ times the accepted reduced bracket $\{\Psi, \Psi^\dagger\}_D = -i\delta$, after smearing. Choosing the opposite holomorphic polarization exchanges exterior multiplication and contraction and is recorded as a candidate coordinate.

Internal gauge and spin transformations act on $\mathfrak{f}$ by the dual of the complete WDW-4B three-generation chiral representation. Their derivations extend to Φ_{odd} by the graded Leibniz rule. Every species, generation, color, weak, hypercharge, spin, and chirality index is therefore present. The zero-fermion classical body does not delete the odd carrier.

9.3 Representation-seed theorem

Theorem W5.8 (CCR/CAR representation-seed theorem).

The branch A test algebra carries the exact metric-scalar smeared CCR seed, and the branch B graded test algebra carries the exact even CCR seed together with the chiral half-density CAR seed determined by the WDW-4B Dirac bracket. The seeds include the full gauge and local-Lorentz response. They do not determine a Hilbert completion, a vacuum, polarization equivalence, determinant, Pfaffian, or fermionic measure.

Proof.

The bosonic commutators follow from the Fréchet product rule:

$$d(\lambda(\cdot)F)[v] = \lambda(v)F + \lambda(\cdot)dF[v].$$

Substitution into the commutator gives $i\hbar\lambda(v)$; two multiplications commute and two constant-direction derivatives commute. The normalization agrees with Theorem W5.4 in every canonical sector.

On the exterior module, the standard identity $\iota_u(\eta \wedge M) + \eta \wedge \iota_u M = \eta(u)M$ proves the mixed CAR, and graded antisymmetry proves the remaining two relations. Multiplication by $\hbar$ matches $i\hbar\{\Psi, \Psi^\dagger\}_D$. Extending the finite-dimensional fiber representations by the graded Leibniz rule gives the complete gauge and Lorentz response. No topology or scalar product on the exterior module has entered the calculation. $\square$

10. Linear quantum-constraint candidates

10.1 Smearing classes and derivations

The load-bearing algebra uses smooth smearings,

$$\xi \in \Gamma^\infty(T\Sigma), \quad \lambda \in \Gamma^\infty(\text{ad } P_0), \quad \vartheta = \tfrac{1}{2}\vartheta^{IJ}J_{IJ} \in \Gamma^\infty(\mathfrak{spin}(1,3)), \quad \vartheta^{IJ} = -\vartheta^{JI},$$

and later $N \in C^\infty(\Sigma)$. Smoothness is stronger than the minimum Sobolev threshold and makes every displayed action well typed. Compactness and $\partial\Sigma = \emptyset$ remove surface terms but do not erase large or improper transformations from the guard ledger. The six independent ϑ^{IJ} contain the three boosts J_{0a} and the three rotations J_{ab} ; no time gauge or $\mathfrak{spin}(3)$ reduction is taken. This smearing is distinct from the inherited QCD coefficient θ .

Let $\mathcal{L}_X$ denote the derivative of a test functional along a field-space fundamental vector field X , including the induced graded action on exterior generators. Define

$$\widehat{\mathcal{D}}[\xi] = -i\hbar \mathcal{L}_{X_\xi}, \quad \widehat{\mathcal{G}}[\lambda] = -i\hbar \mathcal{L}_{X_\lambda}, \quad \widehat{\mathcal{J}}[\vartheta] = -i\hbar \mathcal{L}_{X_\vartheta}.$$

Only $\widehat{\mathcal{D}}$ occurs on branch A . All three occur on branch B .

On configuration generators their actions are:

$$\begin{aligned} \widehat{\mathcal{D}}[\xi] q &= -i\hbar \mathcal{L}_\xi q, \\ \widehat{\mathcal{G}}[\lambda] \mathcal{A} &= +i\hbar D_{\mathcal{A}} \lambda, & \widehat{\mathcal{G}}[\lambda] H &= -i\hbar \rho_{H^*}(\lambda) H, \\ \widehat{\mathcal{G}}[\lambda] \Psi &= -i\hbar \rho_{\text{SM}^*}(\lambda) \Psi, \\ \widehat{\mathcal{J}}[\vartheta] e_i^I &= -i\hbar \vartheta^I{}_J e_i^J, & \widehat{\mathcal{J}}[\vartheta] \Psi &= -i\hbar \rho_{\text{spin}^*}(\vartheta) \Psi. \end{aligned}$$

The contragredient actions on $H^\dagger$ and $\Psi^\dagger$, and the adjoint actions on gauge probes, are fixed by the corresponding dual representations. Thus every one of the six inherited Lorentz generators $\mathcal{J}_{IJ}$ acts on the coframe and chiral-spinor sectors, and no Standard-Model sector is omitted.

10.2 Raw and improved spatial conventions

The accepted classical relation is

$$\mathcal{D}[\xi] = \mathcal{V}_{\text{raw}}[\xi] - \mathcal{G}[\iota_\xi \mathcal{A}] - \mathcal{J}[\iota_\xi \omega_{\text{LC}}(e)].$$

For a declared ordering $\mathfrak{o}_{\text{lin}}$ of the field-dependent smearings, define the raw candidate by

$$\widehat{\mathcal{V}}_{\text{raw}, \mathfrak{o}_{\text{lin}}}[\xi] \stackrel{\text{def}}{=} \widehat{\mathcal{D}}[\xi] + \mathfrak{D}_{\mathfrak{o}_{\text{lin}}} \widehat{\mathcal{G}}[\iota_\xi \mathcal{A}] + \mathfrak{D}_{\mathfrak{o}_{\text{lin}}} \widehat{\mathcal{J}}[\iota_\xi \omega_{\text{LC}}(e)].$$

Because $\omega_{\text{LC}}(e)$ is $\mathfrak{spin}(1,3)$ -valued, $\iota_\xi \omega_{\text{LC}}(e)$ lies in the same six-direction smearing space just declared. The correction is therefore typed without a projection or time-gauge truncation. Thus the exact off-shell candidate relation remains **D plus G plus J**. The pure- $\mathcal{D}$ form is available only after the transposed Gauss and Lorentz constraints annihilate a formal solution. No Hamiltonian commutator is used in this identity.

10.3 Infinitesimal algebra

Because these candidates are derivatives of groupoid actions,

$$[\widehat{\mathcal{D}}[\xi], \widehat{\mathcal{D}}[\eta]] = -i\hbar \widehat{\mathcal{D}}[[\xi, \eta]]$$

on the algebraic core, with analogous Lie-algebra relations for $\widehat{\mathcal{G}}$ and $\widehat{\mathcal{J}}$, and the expected semidirect spatial action on their smearings. This is an algebra of the linear infinitesimal generators only. It is not the quantum hypersurface-deformation algebra and says nothing about commutators involving $\widehat{\mathcal{H}}_\perp$.

10.4 Linear-constraint theorem

Theorem W5.9 (Linear quantum-constraint candidate theorem).

The proper spatial, internal-gauge, and local-Lorentz actions define smeared graded derivations on the branch test algebras in the exact inherited normalization. On branch B , the internal and Lorentz candidates act on every gauge, coframe, Higgs, and chiral-fermion sector, and the Lorentz candidate retains all six $\mathfrak{spin}(1,3)$ directions. The complete off-shell raw/improved relation is retained as **D plus G plus J** with an $\mathfrak{spin}(1,3)$ -valued connection correction, while pure $\mathcal{D}$ is only a joint-constraint-surface identity.

Proof.

Theorem W5.3 supplies the differentiable groupoid actions and their fundamental fields. Theorem W5.5 makes differentiation along those fields well defined on the test algebra, and Theorem W5.8 supplies the graded action. Multiplication by $-i\hbar$ fixes the canonical normalization. The sector formulas follow by differentiating the explicit spatial, gauge, and spin transformations. Antisymmetry of ϑ^{IJ} gives six independent generators, and differentiating the $\text{Spin}^+(1,3)$ action retains both the boost and rotation components on e_i^I and every chiral spinor.

The raw candidate is defined by transporting the exact accepted classical improved-generator identity with its two field-dependent corrections and a recorded ordering. Rearrangement gives the displayed D plus G plus J formula identically on the test algebra. The derived spin connection and its contraction are valued in the same full Lorentz algebra as $\hat{\mathcal{J}}$. Removing the last two summands is licensed only on dual elements annihilated by their transpose actions. Exponentiation, unitarity, essential self-adjointness, large-gauge invariance, and commutators with the Hamiltonian candidate are not used. $\square$

11. Regulator-indexed Hamiltonian candidates

11.1 Ordering coordinates

An ordering coordinate is not a word such as “covariant ordering.” For each branch it is the finite ledger

$$\mathfrak{o}_b = (\prec_b, \beta_b, \nabla_b^{\text{conf}}, \mathfrak{o}_{\text{Higgs}}, \mathfrak{o}_{\text{ferm}}, \mathfrak{o}_{\text{dens}}),$$

where:

- $\prec_b$ records the order of coefficient multiplication, heat smoothing, and Fréchet differentiation;
- β_b records any declared first-order term produced by differentiating the kinetic coefficient;
- ∇_b^{conf} is a declared smooth connection on the local affine chart, with the flat chart connection as the baseline;
- $\mathfrak{o}_{\text{Higgs}}$ fixes the order of the independent $H, H^\dagger$ derivatives;
- $\mathfrak{o}_{\text{ferm}}$ fixes exterior multiplication versus contraction in each bilinear; and
- $\mathfrak{o}_{\text{dens}}$ fixes every reference-to-intrinsic density conversion.

For the baseline left-coefficient ordering,

$$\mathfrak{D}_{\mathfrak{o}_b}^{(0)} (\text{Contr}_{\mathbb{K}_b, K_\epsilon} d^2) F = \text{Contr}_{\mathbb{K}_b, K_\epsilon} d^2 F.$$

An admitted alternative has the same regulated second-order principal part and a displayed finite first-order correction

$$\mathfrak{D}_{\mathfrak{o}_b} (\text{Contr}_{\mathbb{K}_b, K_\epsilon} d^2) F = \text{Contr}_{\mathbb{K}_b, K_\epsilon} d^2 F + \mathcal{B}_{b, \epsilon, \mathfrak{o}_b}(q)[dF],$$

where $\mathcal{B}$ is a smooth, explicitly typed contraction of dF . No ordering is identified with another before the intertwiner test of Section 14.

11.2 Core Hamiltonian

Write

$$\mathcal{V}_A(q_A) = -\frac{\sqrt{\hbar}}{2\kappa}(R(h) - 2\Lambda) + \frac{\sqrt{\hbar}}{2}(D_i \phi D^i \phi + m^2 \phi^2).$$

For $N \in C^\infty(\Sigma)$, $\epsilon > 0$, and $\mathfrak{o}_A \in \mathfrak{D}_A$, define

$$\begin{aligned} \widehat{\mathcal{H}}_{\perp, \epsilon, \mathfrak{o}_A}^A[N] F &\stackrel{\text{def}}{=} -\hbar^2 \mathfrak{D}_{\mathfrak{o}_A} (\text{Contr}_{\mathbb{K}_A, K_\epsilon} d^2) F[N] \\ &\quad + \left(\int_\Sigma N \mathcal{V}_A(q_A) \right) F. \end{aligned}$$

The scalar momentum block $1/(2\sqrt{\hbar})$ is already inside $\mathbb{K}_A$; it is not added a second time. The scalar remains constrained matter. No external time derivative is introduced.

In a formal kernel display, the gravitational principal part is the regulated form of

$$-2\kappa\hbar^2 \int_\Sigma N(x) \mathcal{G}_{ijkl}(h; x) \frac{\delta^2}{\delta h_{ij}(x) \delta h_{kl}(x)},$$

and the clock principal part is the regulated form of

$$-\frac{\hbar^2}{2} \int_\Sigma \frac{N(x)}{\sqrt{\hbar}(x)} \frac{\delta^2}{\delta \phi(x)^2}.$$

These two lines are mnemonic consequences of the heat-kernel trace, not definitions of coincident products.

11.3 Standard-Model extension

The complete even potential density is

$$\begin{aligned} \mathcal{V}_B = & -\frac{\sqrt{h}}{2\kappa}(R(h) - 2\Lambda) + \frac{\sqrt{h}}{2}(D_i\phi D^i\phi + m^2\phi^2) \\ & + \sum_{r=3,2,Y} \frac{\sqrt{h}}{4} F_{r,ij} \cdot F_r{}^{ij} \\ & + \sqrt{h}(h^{ij}(\mathcal{D}_i H)^\dagger \mathcal{D}_j H + V_H(H)). \end{aligned}$$

Its conjugate momenta are already included in $\mathbb{K}_B$:

$$\frac{p_\phi^2}{2\sqrt{h}}, \quad \sum_r \frac{E_r{}^i \cdot E_{r,i}}{2\sqrt{h}}, \quad \frac{\Pi_H \Pi_H^\dagger}{\sqrt{h}}.$$

Let $\mathcal{D}_i(e, \mathcal{A})$ contain the derived $\omega_{\text{LC}}(e)$, the three internal gauge connections, and the half-density connection. Put

$$\mathcal{S}_\infty = \bigcap_{k \geq 0} \mathcal{S}_{\text{SM}, \text{test}}^{s+k}, \quad p_k(u) = \|u\|_{H^{s+k}}.$$

The lapse-smearred Dirac and Yukawa one-particle maps are

$$A_{D,N}(q), A_{Y,N}(q) : \mathcal{S}_\infty \longrightarrow \mathcal{S}_\infty.$$

They include every species, generation, color, weak, hypercharge, chirality, and spin index inherited from WDW-4B. Locally on the smooth configuration core they obey

$$\begin{aligned} p_k(A_{D,N}(q)u) & \leq C_{k,q,N} p_{k+1}(u), \\ p_k(A_{Y,N}(q)u) & \leq C_{k,q,N} p_k(u). \end{aligned}$$

For every configuration derivative and tangent tuple $\mathbf{v} = (v_1, \dots, v_\ell)$, the corresponding estimates are

$$\begin{aligned} p_k(d^\ell A_{D,N}(q)[\mathbf{v}]u) & \leq C_{k,\ell,q,N} \prod_a \|v_a\|_{H^{r_{k,\ell}}} p_{k+1}(u), \\ p_k(d^\ell A_{Y,N}(q)[\mathbf{v}]u) & \leq C'_{k,\ell,q,N} \prod_a \|v_a\|_{H^{r'_{k,\ell}}} p_k(u). \end{aligned}$$

The coefficients are smooth functions of the coframe, derived spin connection, three gauge connections, Higgs field, lapse, and density data; Sobolev multiplication and the first-order differential estimate prove these bounds. Thus $q \mapsto A_{D,N}(q), A_{Y,N}(q)$ is Bastiani smooth in the bounded-operator topology of the Fréchet scale.

Let $A_N(q) = A_{D,N}(q) + A_{Y,N}(q)$. On the degreewise complete exterior-section carrier defined in Section 12, the ordered fermion action is

$$\begin{aligned} (\widehat{\mathcal{H}}_{\Psi, \mathfrak{o}_{\text{ferm}}}[N]F)^{(n)}(q) & \stackrel{\text{def}}{=} d\Gamma_n(A_N(q)')F^{(n)}(q) \\ & + \sum_{\delta \in \Delta_{\mathfrak{o}}} R_{\delta, \mathfrak{o}}^{(n)}(q)F^{(n)}(q), \end{aligned}$$

where $A_N(q)'$ is the strong transpose, $d\Gamma_n$ is its continuous degree- n exterior lift, and each recorded

$$R_{\delta, \mathfrak{o}}^{(n)}(q) : \mathfrak{F}_B^{(n)} \longrightarrow \mathfrak{F}_B^{(n+\delta)} \quad (\delta \in \{-2, 0, 2\})$$

is an explicitly typed finite composition of exterior multiplication and smooth-core contraction. The baseline holomorphic ordering has $\Delta_{\mathfrak{o}} = \emptyset$ and is degree preserving. An alternative ordering is admitted only when its finite list of degree shifts and any scalar term $c_{\mathfrak{o}}(q)\mathbf{1}$ are printed in $\mathfrak{o}_{\text{ferm}}$ and are continuous on every affected completed degree. No normal-ordering constant is inserted implicitly.

The Yukawa map uses the fixed matrices Y_u, Y_d, Y_e and the one accepted Higgs doublet. The local θ coordinate is carried through the accepted canonical convention; on the fixed trivial closed branch it adds no unregistered independent energy density.

The branch B Hamiltonian candidate is

$$\begin{aligned} \widehat{\mathcal{H}}_{\perp, \epsilon, \mathfrak{o}_B}^B[N]F & \stackrel{\text{def}}{=} -\hbar^2 \mathfrak{D}_{\mathfrak{o}_B} (\text{Contr}_{\mathbb{K}_B, K_\epsilon} d^2) F[N] \\ & + \left(\int_{\Sigma} N \mathcal{V}_B \right) F + \widehat{\mathcal{H}}_{\Psi, \mathfrak{o}_{\text{ferm}}}[N]F. \end{aligned}$$

Every occurrence of the spin connection in this formula means $\omega_{\text{LC}}(e)$. No independent connection derivative or Einstein-Cartan variable is inserted.

11.4 Source, density, and branch audit

The complete candidate family is

$$\begin{aligned}\widehat{\mathcal{C}}_{5A,\epsilon,\mathfrak{o}_A} &= (\widehat{\mathcal{D}}, \widehat{\mathcal{H}}_{\perp,\epsilon,\mathfrak{o}_A}^A), \\ \widehat{\mathcal{C}}_{5B,\epsilon,\mathfrak{o}_B} &= (\widehat{\mathcal{D}}, \widehat{\mathcal{G}}, \widehat{\mathcal{J}}, \widehat{\mathcal{H}}_{\perp,\epsilon,\mathfrak{o}_B}^B).\end{aligned}$$

All entries are smeared. The exact fixed parameters

$$(\kappa, \Lambda, m, g_3, g_2, g_1, Y_u, Y_d, Y_e, V_H, \theta)$$

are coefficients, not derived outputs. The core and extension families have different configuration variables and are not set equal.

11.5 Hamiltonian theorem

Theorem W5.10 (Regulator-indexed Hamiltonian candidate theorem).

For every $\epsilon > 0$, allowed lapse N , and declared ordering $\mathfrak{o}_b$, the displayed formulas define graded algebraic actions on the corresponding smooth cylinder/local class. The branch- A action is a total endomorphism of its scalar Bastiani algebra. The branch- B action, including the complete Dirac/Yukawa CAR term and the section-valued regulated Hessian, is a total endomorphism of the degreewise complete finite-particle section algebra $\mathcal{E}_{5B}^{\text{sec}}$ defined below. They contain:

1. the exact WDW-1 gravitational normalization, DeWitt sign, and cosmological convention;
2. the complete scalar-clock kinetic and potential terms;
3. on branch B , all gauge electric/magnetic, Higgs momentum/gradient/potential, and chiral fermion/Yukawa terms;
4. the exact density and independent-complex-Higgs conventions;
5. only the derived torsion-free spin connection; and
6. explicit regulator, ordering, branch, and fixed-target coordinates.

Proof.

The scalar part of Theorem W5.7 makes each branch- A second-order contraction finite and typed for $\epsilon > 0$. Its complete-target extension applies degreewise to branch B . Multiplication by the displayed local potentials is defined because $s > 7/2$ gives C^2 control and a Sobolev algebra, while curvature has the recorded H^{s-2} type.

The scale estimates above prove that $A_{D,N}(q)$ and $A_{Y,N}(q)$ are continuous endomorphisms of $\mathcal{S}_\infty$ with smooth configuration dependence. Section 12 proves that their strong transposes act continuously on $\mathfrak{f}_\infty$, that $d\Gamma_n(A_N(q)')$ extends continuously to every completed exterior degree, and that the recorded finite ordering maps have the stated degreewise types. Therefore the complete fermionic action is total on $\mathcal{E}_{5B}^{\text{sec}}$, including on sections with infinite-dimensional range inside one exterior degree.

Polarizing the exact classical quadratic momentum terms gives $\mathbb{K}_A$ and $\mathbb{K}_B$, so no coefficient is guessed. The remaining displayed terms reproduce the full WDW-4A and WDW-4B potential ledgers. The ordering tuple records every permitted rearrangement and its first-order correction. Compactness removes surface integrals, while the target guard excludes boundary completions. Thus every term is a total algebraic action on the corresponding $\mathcal{E}_{5b}^{\text{sec}}$ with all indices visible. The proof contains no assertion about a limit as $\epsilon \downarrow 0$, an adjoint, or a Hamiltonian commutator. $\square$

12. The common algebraic action domain

12.1 Primitive-stable ambient algebra

The smooth spinor core is the nuclear Fréchet space

$$\mathcal{S}_\infty = \bigcap_{k \geq 0} \mathcal{S}_{\text{SM},\text{test}}^{s+k}.$$

Its strong dual is the complete nuclear DF -space

$$\mathfrak{f}_\infty = (\mathcal{S}_\infty)'_b \cong \varinjlim_{k \geq 0} (\mathcal{S}_{\text{SM},\text{test}}^{s+k})^*.$$

For $n \geq 0$, define the degreewise completed projective exterior carrier

$$\mathfrak{F}_B^{(n)} = \widehat{\Lambda}_\pi^n \mathfrak{f}_\infty, \quad \mathfrak{F}_B^{(0)} = \mathbb{C},$$

and the algebraic finite-particle sum

$$\mathfrak{F}_{B,\text{fin}} = \bigcup_{N \geq 0} \bigoplus_{n=0}^N \mathfrak{F}_B^{(n)}.$$

The completion is only within a fixed exterior degree. There is no completion across degrees, Fock norm, vacuum, scalar product, determinant, Pfaffian, or fermionic measure.

On the smooth inverse-limit configuration core define

$$\mathcal{E}_{5B}^{\text{sec}} = \bigcup_{N \geq 0} \bigoplus_{n=0}^N C_B^\infty(\mathcal{U}_{5B}^\infty; \mathfrak{F}_B^{(n)}).$$

The wedge product is the continuous map induced by the completed projective tensor product and makes this an algebra because every element has finite particle degree. Branch A uses

$$\mathcal{E}_{5A}^{\text{sec}} = C_B^\infty(\mathcal{U}_{5A}^\infty; \mathbb{C}).$$

Write $\mathcal{E}_{5b}^{\text{sec}}$ for the selected branch.

We now assign the topology used later by candidate comparison. Let $\mathcal{X}_b^\infty$ be the Fréchet model space of $\mathcal{U}_{5b}^\infty$, let

$$Y_{A,0} = \mathbb{C}, \quad Y_{B,n} = \mathfrak{F}_B^{(n)},$$

and give each target its displayed locally convex topology. For $F \in C_B^\infty(\mathcal{U}_{5b}^\infty; Y_{b,n})$, $\ell \geq 0$, a continuous seminorm p on $Y_{b,n}$, and a compact set

$$K \subset \mathcal{U}_{5b}^\infty \times (\mathcal{X}_b^\infty)^\ell,$$

set

$$\|F\|_{\ell,K,p} = \sup_{(q;v_1,\dots,v_\ell) \in K} p(d^\ell F(q)[v_1, \dots, v_\ell]).$$

For $\ell = 0$, the direction factor is omitted. These seminorms are finite because the displayed derivative is continuous on the compact set K ; they separate sections already at $\ell = 0$. They define the Hausdorff compact-open Bastiani topology $\tau_{b,n}$ on each degree. The branch B ambient carrier receives the locally convex direct-sum topology

$$\tau_{B,\text{sec}} = \bigoplus_{n \geq 0} \tau_{B,n},$$

and branch A receives $\tau_{A,\text{sec}} = \tau_{A,0}$. Thus the displayed union is topologized without completing it across particle degree.

The topology is compatible with the algebra and every ambient action used below. On fixed degrees the target wedge

$$\mathfrak{F}_B^{(n)} \times \mathfrak{F}_B^{(m)} \longrightarrow \mathfrak{F}_B^{(n+m)}$$

is continuous. The Bastiani Leibniz rule expresses each derivative of the pointwise wedge as a finite sum of continuous target wedges; projection of a compact derivative-test set is compact. Hence the pointwise degree (n, m) product is continuous for $\tau_{B,n} \times \tau_{B,m} \rightarrow \tau_{B,n+m}$. The universal property of the projective tensor product and the canonical direct-sum identity

$$\left(\bigoplus_n E_n \right) \otimes_\pi \left(\bigoplus_m E_m \right) \cong \bigoplus_{n,m} (E_n \otimes_\pi E_m)$$

for the degreewise mapping spaces E_n reduce continuity to the degreewise maps. Together with the finite sum $\sum_{n+m=k}$ in each output degree k , this makes the graded wedge product jointly continuous on $(\mathcal{E}_{5B}^{\text{sec}}, \tau_{B,\text{sec}})$. Constant-direction differentiation merely raises ℓ in the defining seminorms, and multiplication by a fixed smooth coefficient is continuous by the same Leibniz estimate. Smooth proper-redescription pullbacks are continuous because compact derivative-test sets have compact images and the degree actions are continuous.

This definition deliberately admits vector-valued smooth sections with infinite-dimensional range inside one degree. The former fixed-monomial class

$$\left\{ \sum_{j=1}^m F_j(q) M_j : M_j \text{ fixed} \right\}$$

is only the finite-rank constant-section subalgebra; it is not the ambient branch- B carrier.

The restriction map

$$j_s : (\mathcal{S}_{\text{SM,test}}^s)^* \longrightarrow \mathfrak{f}_\infty$$

is injective because $\mathcal{S}_\infty$ is dense in the fixed H^s test space. Hence the induced algebraic exterior map

$$\Lambda_{\text{alg}}(j_s) : \Lambda_{\text{alg}}((\mathcal{S}_{\text{SM},\text{test}}^s)^*) \hookrightarrow \mathfrak{F}_{B,\text{fin}}$$

is injective and embeds the W5.8 CAR seed as constant sections. On the full strong-dual carrier, contraction is taken only against $u \in \mathcal{S}_\infty$, for which $\eta \mapsto \eta(u)$ is continuous. The fixed-rung H^s contractions remain valid on the embedded seed. Constraint primitives use smooth spinor smearings, so this totality restriction removes no declared constraint direction.

We next prove the one-particle action before forming the common domain. For a bounded set $B \subset \mathcal{S}_\infty$, let

$$p_B(\eta) = \sup_{u \in B} |\eta(u)|$$

be the corresponding strong-dual seminorm. The scale estimates in Section 11 give

$$p_B(A_N(q)'\eta) = p_{A_N(q)B}(\eta).$$

Continuous endomorphisms take bounded sets to bounded sets; hence

$$A_N(q)' : \mathfrak{f}_\infty \longrightarrow \mathfrak{f}_\infty$$

is continuous. Moreover,

$$d^\ell(A_N(\cdot)')(q)[\mathbf{v}]\eta = (d^\ell A_N(q)[\mathbf{v}])'\eta,$$

and the rungwise estimates show locally and for every bounded B that the right side is controlled by one strong-dual seminorm. Thus $q \mapsto A_N(q)'$ is Bastiani smooth in the bounded-convergence operator topology; smoothness has not been inferred from transposition alone.

For each n , define on simple wedges

$$d\Gamma_n(A_N(q)') \stackrel{\text{def}}{=} \text{Alt}_n \sum_{\ell=1}^n 1^{\widehat{\otimes}(\ell-1)} \widehat{\otimes} A_N(q)' \widehat{\otimes} 1^{\widehat{\otimes}(n-\ell)}.$$

The universal property of the projective tensor product and continuity of antisymmetrization extend this uniquely to

$$d\Gamma_n(A_N(q)') : \mathfrak{F}_B^{(n)} \longrightarrow \mathfrak{F}_B^{(n)}.$$

The bounded-set estimates above prove smooth dependence on q . Consequently

$$F^{(n)}(q) \longmapsto d\Gamma_n(A_N(q)')F^{(n)}(q)$$

maps smooth degree- n sections to smooth degree- n sections. The same projective-tensor argument types every finite ordering map $R_{\delta,\sigma}^{(n)}(q)$; admissibility of $\mathfrak{o}_{\text{ferm}}$ requires continuity and smooth parameter dependence on each affected degree.

The regulated Hessian also acts on the entire section carrier. Each $\mathfrak{F}_B^{(n)}$ is complete. Continuity of a Bastiani derivative at q gives, for every target seminorm, a neighborhood and a finite Sobolev rung controlling that derivative. The complete-target estimate in Theorem W5.7 therefore gives

$$\sum_k p(d^2 F^{(n)}(q) [S_{\epsilon,b}^{(r)} e_k, T_{\epsilon,b}^{(r)} e_k]) \leq C_{p,q} \|S_{\epsilon,b}^{(r)}\|_2 \|T_{\epsilon,b}^{(r)}\|_2.$$

The sum converges in $\mathfrak{F}_B^{(n)}$; the differentiated estimates prove smoothness. Thus a later Hamiltonian may act on the output of the fermionic Hamiltonian without forcing that output into a finite-dimensional monomial span.

The spatial, gauge, and Lorentz actions lift similarly. Their infinitesimal action on a section is the sum of the base derivative $dF[X]$ and the degreewise exterior lift of the smooth one-particle dual action. The one-rung spatial loss is absorbed by $\mathcal{S}_\infty$ and the inverse-limit configuration core. All six Lorentz directions act on the same carrier.

Fix a branch b , one smoothing family $\mathcal{K}_b$, one ordering ledger $\mathfrak{o}_b$, and the full range $\epsilon > 0$ and allowed smearings. Let $\mathcal{P}_b$ contain:

- configuration multiplication and constant smooth-core Fréchet derivatives;
- continuous exterior multiplication and smooth-core contraction;
- the lifted spatial, gauge, and full six-direction Lorentz derivations present on the branch;
- multiplication by every inherited local potential coefficient;
- every degreewise $d\Gamma_n(A_N(q)')$ and every explicitly recorded finite fermionic ordering component;

- the scalar or section-valued regulated Hessian traces for every $\epsilon > 0$; and
- the finite first-order ordering corrections in $\mathfrak{o}_b$.

The preceding constructions prove separately, before Φ_{5b} is defined, that every primitive is a total linear map

$$P : \mathcal{E}_{5b}^{\text{sec}} \longrightarrow \mathcal{E}_{5b}^{\text{sec}}.$$

In particular, for fixed $\eta \in \mathfrak{f}_\infty$, the map

$$q \mapsto A_N(q)' \eta$$

is a legitimate smooth degree-one section even when its range is infinite-dimensional. This is the totality fact required by the complete fermionic Hamiltonian, not a status assigned by the later domain recursion.

The same estimates also prove continuity for the topology just declared. For a smooth family of continuous one-particle maps, the differentiated product rule reduces every $\|\cdot\|_{\ell,K,p}$ seminorm of $q \mapsto d\Gamma_n(A_N(q)')F^{(n)}(q)$ to finitely many input seminorms. Exterior multiplication and smooth-core contraction are continuous degreewise. The complete-target trace estimate, uniformly over each compact derivative-test set, bounds the regulated Hessian map by finitely many input seminorms two derivative orders higher. Multiplication and first-order ordering terms were covered above. Consequently every $P \in \mathcal{P}_b$ is not only total but continuous:

$$P : (\mathcal{E}_{5b}^{\text{sec}}, \tau_{b,\text{sec}}) \longrightarrow (\mathcal{E}_{5b}^{\text{sec}}, \tau_{b,\text{sec}}).$$

This locally convex statement is internal to the WDW-5 comparison interface; it is not a Hilbert topology, a closed-operator assertion, or a completion of Φ_{5b} .

The initial algebras embed as

$$\mathcal{A}_b^{(0)} \subset \mathcal{E}_{5b}^{\text{sec}}.$$

Cylinder and local functions are Bastiani smooth, and the W5.8 exterior polynomials embed through $\Lambda_{\text{alg}}(j_s)$. The Hilbert-chart CCR/CAR seed remains defined for all declared fixed-rung test directions on this initial subalgebra, while the total primitive family on the enlarged carrier uses the dense smooth-direction subfamily required by the constraints.

For clarity, these are two typed representation levels, not one action with an implicit domain change. Let $\mathfrak{R}_{s,b}$ be the full fixed-rung CCR/CAR seed algebra of Theorem W5.8 and write

$$\rho_{s,b}^{(0)} : \mathfrak{R}_{s,b} \longrightarrow \text{End}(\mathcal{A}_b^{(0)})$$

for exactly that action. Let $\mathfrak{R}_{\infty,b}$ be generated by configuration multiplications, momentum derivatives in $\mathcal{X}_b^\infty$, and, on branch B , exterior multiplication by $\eta \in \mathfrak{f}_\infty$ and contraction by $u \in \mathcal{S}_\infty$. The constructions above give a continuous smooth-core action

$$\rho_{\infty,b} : \mathfrak{R}_{\infty,b} \longrightarrow \text{End}_{\text{cont}}(\mathcal{E}_{5b}^{\text{sec}}, \tau_{b,\text{sec}}).$$

On the common directions $v \in \mathcal{X}_b^\infty$, $\eta = j_s(\eta_s)$, and $u \in \mathcal{S}_\infty$, the two formulas agree on $\mathcal{A}_b^{(0)}$. No action of a nonsmooth $v \in \mathcal{H}_b^s \setminus \mathcal{X}_b^\infty$ or $u \in \mathcal{S}_{\text{SM,test}}^s \setminus \mathcal{S}_\infty$ is asserted on the enlarged carrier. In particular, embedding the fixed-rung seed does not falsely extend all of its operators to $\mathcal{E}_{5b}^{\text{sec}}$.

12.2 Unguarded finite-primitive closure

Set

$$\Phi_{5b}^{(0)} = \text{Alg}_{\mathbb{C}}^{\text{gr}}(\mathcal{A}_b^{(0)}) \subset \mathcal{E}_{5b}^{\text{sec}}.$$

Recursively, let $\Phi_{5b}^{(n+1)}$ be the unital graded algebraic span of $\Phi_{5b}^{(n)}$ and all

$$P(F), \quad F \in \Phi_{5b}^{(n)}, \quad P \in \mathcal{P}_b,$$

with no domain guard or “for which defined” qualifier. Totality on $\mathcal{E}_{5b}^{\text{sec}}$ proves that every displayed $P(F)$ exists and remains in $\mathcal{E}_{5b}^{\text{sec}}$. Define

$$\Phi_{5b} = \bigcup_{n=0}^{\infty} \Phi_{5b}^{(n)}.$$

This is a set-theoretic algebraic union, not a norm completion. Each element has a semantic value in $\mathcal{E}_{5b}^{\text{sec}}$ and is produced by a finite word in total, explicitly constructed primitives. Induction on word length is therefore a semantic evaluation theorem: the empty word evaluates in $\Phi^{(0)}$; if a word w evaluates in $\Phi^{(n)}$, then $P \circ w$ evaluates in $\Phi^{(n+1)}$ because $P : \mathcal{E}_{5b}^{\text{sec}} \rightarrow \mathcal{E}_{5b}^{\text{sec}}$ is total. Heat smoothing remains in every second-order contraction, so iteration does not create an unregulated diagonal.

For a candidate constraint C and $F \in \Phi_{5b}^{(n)}$, every term of $\widehat{C}F$ is one of the primitive images used to form $\Phi_{5b}^{(n+1)}$. Hence

$$\widehat{C}_{\epsilon,0}[\alpha] : \Phi_{5b} \longrightarrow \Phi_{5b}.$$

The same domain works for every smearing and every positive regulator in the fixed candidate family.

Every finite composition of individual primitives or constraints consequently exists algebraically and maps Φ_{5b} to itself. This single-action and finite-word fact does not establish a common dense Hilbert domain, graph closure, adjoints, strong or weak convergence of regulated commutators, or closure of the resulting commutator expressions in the classical constraint form.

Because every generator of $\rho_{\infty,b}$ belongs to $\mathcal{P}_b$, the same induction gives the separately typed invariant action

$$\rho_{\infty,b}(a) : \Phi_{5b} \longrightarrow \Phi_{5b}, \quad a \in \mathfrak{R}_{\infty,b}.$$

The full fixed-rung action $\rho_{s,b}^{(0)}$ remains, exactly as defined, on the initial seed subalgebra $\mathcal{A}_b^{(0)}$.

12.3 Algebraic transpose

Define the full algebraic dual

$$\Phi_{5b}^\times = \text{Hom}_{\mathbb{C}}(\Phi_{5b}, \mathbb{C}).$$

For every constraint action, its transpose is

$$\begin{aligned} \widehat{C}_{\epsilon,0}^\times[\alpha] : \Phi_{5b}^\times &\longrightarrow \Phi_{5b}^\times, \\ (\widehat{C}_{\epsilon,0}^\times[\alpha]\Psi)(F) &\stackrel{\text{def}}{=} \Psi(\widehat{C}_{\epsilon,0}[\alpha]F). \end{aligned}$$

The formula is typed precisely because $\widehat{C}_{\epsilon,0}[\alpha]F \in \Phi_{5b}$. No continuity of Ψ or of the transpose is claimed.

12.4 Common-domain theorem

Theorem W5.11 (Common algebraic action-domain theorem).

For each complete candidate coordinate, every declared primitive is first a continuous total endomorphism of the concretely topologized section algebra $(\mathcal{E}_{5b}^{\text{sec}}, \tau_{b,\text{sec}})$. The unguarded inductive construction then gives one graded algebra $\Phi_{5b} \subset \mathcal{E}_{5b}^{\text{sec}}$ such that every smeared, regulated constraint acts as

$$\widehat{C}_{\epsilon,0}[\alpha] : \Phi_5 \longrightarrow \Phi_5$$

(with $\Phi_5 = \Phi_{5b}$ on the selected branch). Its transpose acts on the full algebraic dual as

$$\widehat{C}_{\epsilon,0}^\times[\alpha] : \Phi_5^\times \longrightarrow \Phi_5^\times.$$

The smooth-core representation $\rho_{\infty,b}$ preserves Φ_{5b} ; the full fixed-rung representation $\rho_{s,b}^{(0)}$ remains separately typed on $\mathcal{A}_b^{(0)}$. The construction is noncircular and does not imply any Hilbert-space density, Hilbert-domain invariance, adjoint, closability, regulator removal, or quantum commutator closure.

Proof.

The base algebra and the ambient Bastiani-smooth section algebra are constructed before any constraint closure. The primitive actions are explicit formulas independently typed in Theorems W5.5–W5.10 and Section 12.1. The strong-dual estimates, completed projective exterior lifts, vector-field rules, and complete-target trace estimate prove separately that the Dirac/Yukawa action, every finite ordering component, every linear action, and every regulated Hessian contraction are total on $\mathcal{E}_{5b}^{\text{sec}}$. The recursive definition therefore uses no partial application and defines $\Phi^{(n+1)}$ without presupposing the final invariant domain.

If $F \in \Phi_5$, then $F \in \Phi^{(n)}$ for some finite n . Every term in $\widehat{C}F$ is a primitive image and hence belongs to $\Phi^{(n+1)} \subset \Phi_5$. This proves the single-action map $\Phi_5 \rightarrow \Phi_5$. The same induction proves total semantic evaluation of every finite constraint word. Precomposition with each single-action map sends every linear functional on Φ_5 to another linear functional on Φ_5 , proving the transpose type.

The transpose construction uses Φ_5 only algebraically. Although Φ_5 sits inside the topologized ambient carrier and may carry its subspace topology for candidate comparison, no Hilbert or graph topology, completion, or measure identifies it with a dense operator domain. Algebraic finite-word closure therefore does not settle any graph-domain, adjoint, convergence, renormalization, or anomaly question. $\square$

13. Formal joint solutions in the algebraic dual

13.1 Branch solution functors

Let

$$\alpha_b = (\mathcal{K}_b, \epsilon, \mathfrak{o}_b, \rho_b, \mathcal{P}_b)$$

denote a complete variable candidate coordinate at the fixed source and target. For branch A , define

$$\begin{aligned} \text{Sol}_{5A}(\mathbf{Q}_{5A, \alpha_A}) = \{ \Psi \in \Phi_{5A}^\times : \widehat{\mathcal{D}}^\times[\xi]\Psi = 0 \quad \forall \xi, \\ (\widehat{\mathcal{H}}_{\perp, \epsilon, \mathfrak{o}_A}^A)^\times[N]\Psi = 0 \quad \forall N \}. \end{aligned}$$

For branch B ,

$$\begin{aligned} \text{Sol}_{5B}(\mathbf{Q}_{5B, \alpha_B}) = \{ \Psi \in \Phi_{5B}^\times : \widehat{\mathcal{D}}^\times[\xi]\Psi = 0 \quad \forall \xi, \\ \widehat{\mathcal{G}}^\times[\lambda]\Psi = 0 \quad \forall \lambda, \\ \widehat{\mathcal{J}}^\times[\vartheta]\Psi = 0 \quad \forall \vartheta = \tfrac{1}{2}\vartheta^{IJ}J_{IJ} \in \Gamma^\infty(\mathfrak{spin}(1, 3)), \\ (\widehat{\mathcal{H}}_{\perp, \epsilon, \mathfrak{o}_B}^B)^\times[N]\Psi = 0 \quad \forall N \}. \end{aligned}$$

Every equation is shorthand for an annihilation identity on all $F \in \Phi_{5b}$. For example,

$$\Psi(\widehat{\mathcal{H}}_{\perp, \epsilon, \mathfrak{o}_b}^b[N]F) = 0 \quad \text{for all } N, F.$$

The zero functional always satisfies these homogeneous identities. No nonzero solution is asserted. The object is called a formal joint annihilator, not $\mathcal{H}_{\text{phys}}$.

13.2 Proper-equivalence predicate

Consider two candidates $\mathbf{Q}, \mathbf{Q}'$ in the same frozen classical target class, with branch labels b, b' . A complete constraint label is not only a smearing symbol. Let

$$\mathfrak{I}(\mathbf{Q}) = \coprod_{\mathfrak{t} \in \{\mathbf{D}, \mathbf{G}, \mathbf{J}, \mathbf{H}\}_b} \mathfrak{I}_{\mathfrak{t}}(\mathbf{Q})$$

be the disjoint union of the branch-appropriate spatial, internal-Gauss, Lorentz, and Hamiltonian labels. An element records

$$i = (b, \mathfrak{t}, \sigma; \mathcal{K}_b, \epsilon, \nabla_b^{\text{conf}}, \mathfrak{o}_b, \mathfrak{o}_{\text{dens}}, \rho_b^{\text{pol}}),$$

with the null symbol in a slot not used by that constraint type. Thus $\widehat{C}_i$ determines one and only one smeared action, including its regulator, smoothing/pivot, density, connection, polarization, and ordering coordinates whenever they are operative.

A **complete candidate matching** $\mathbb{T} : \mathbf{Q} \rightarrow \mathbf{Q}'$ is the tuple

$$\mathbb{T} = (\iota_4, \kappa, \Gamma, U_s, U, U_s^\vee, U^\vee, \mu_s, \mu_\infty, \mu_C, \widetilde{T})$$

with the following typed components:

$$\begin{aligned} \iota_4 : \mathfrak{H}_{4,b}^{\text{cl}} &\longrightarrow \mathfrak{H}_{4,b'}^{\text{cl}}, \\ \kappa : \mathcal{U}_{5b}^\infty &\longrightarrow (\mathcal{U}_{5b'}^\infty)^\infty, \\ \Gamma : \mathcal{G}_{5b}^{\text{prop}} &\longrightarrow (\mathcal{G}_{5b'}^{\text{prop}})^{\text{prop}}, \\ U_s : \mathcal{S}_{\text{SM}, \text{test}}^s &\longrightarrow (\mathcal{S}_{\text{SM}, \text{test}}^s)^s, \quad U : \mathcal{S}_\infty \longrightarrow \mathcal{S}'_\infty, \\ \mu_s : \mathfrak{R}_{s,b} &\longrightarrow \mathfrak{R}'_{s,b'}, \quad \mu_\infty : \mathfrak{R}_{\infty,b} \longrightarrow \mathfrak{R}'_{\infty,b'}, \\ \mu_C : \mathfrak{I}(\mathbf{Q}) &\longrightarrow \mathfrak{I}(\mathbf{Q}'). \end{aligned}$$

Here ι_4 is an isomorphism of the registered branch source, κ is a Bastiani-smooth diffeomorphism, Γ is a proper Lie-groupoid isomorphism over κ , U_s, U are continuous linear isomorphisms with $U_s|_{\mathcal{S}_\infty} = U$, and μ_s, μ_∞ are unital graded algebra isomorphisms. The dual maps are not independent choices:

$$U_s^\vee = (U_s^{-1})', \quad U^\vee = (U^{-1})', \quad U^\vee j_s = j'_s U_s^\vee.$$

They preserve the CAR pairing,

$$(U^\vee \eta)(Uu) = \eta(u),$$

and induce bicontinuous maps $\widehat{\Lambda}_\pi^n U^\vee$ on every completed exterior degree. On branch A these one-particle entries are the unique maps of the trivial degree-zero carrier.

The ambient map is the induced filtered graded algebra isomorphism

$$\widetilde{T} : (\mathcal{E}_{5b}^{\text{sec}}, \tau_{b,\text{sec}}) \longrightarrow ((\mathcal{E}'_{5b'})^{\text{sec}}, \tau'_{b',\text{sec}}),$$

defined on branch B by

$$(\widetilde{T}F)^{(n)}(\kappa(q)) = \widehat{\Lambda}_\pi^n(U^\vee)F^{(n)}(q)$$

and by scalar pullback along κ^{-1} on branch A . It is required to map $\mathcal{A}_b^{(0)}$ onto $(\mathcal{A}'_{b'})^{(0)}$ and Φ_{5b} onto $\Phi'_{5b'}$. Write

$$T = \widetilde{T}|_{\Phi_{5b}}, \quad T_0 = \widetilde{T}|_{\mathcal{A}_b^{(0)}}, \quad T^\times \Psi' = \Psi' \circ T.$$

The tuple is admitted only when the following nine standing-bearing compatibilities hold.

1. **source and branch:** ι_4 commutes with the registered WDW-4 source projections. On a fixed branch it is the identity forced by the frozen source. A cross-branch match is not inferred; it would require an actual ι_4 and is impossible in the baseline family when the nontrivial exterior degrees have no carrier isomorphism.
2. **base and redescription:** Γ covers κ , and

$$T\rho(g) = \rho'(\Gamma g)T \quad \text{for every } g \in \mathcal{G}_{5b}^{\text{prop}}.$$

3. **topological carrier:** $\widetilde{T}$ and its inverse are continuous for the printed compact-open Bastiani/direct-sum topologies and preserve the degree filtration, wedge, smooth-core rigging, strong dual, fixed-rung seed embedding, and initial algebra.
4. **fixed-rung action:** generator transport by $(\kappa, U_s, U_s^\vee)$ defines μ_s , and only on the actual fixed-rung domain

$$T_0\rho_{s,b}^{(0)}(a) = (\rho'_{s,b'})^{(0)}(\mu_s(a))T_0 \quad \forall a \in \mathfrak{R}_{s,b}.$$

5. **smooth-core action:** generator transport by $(\kappa, U, U^\vee)$ defines μ_∞ , agrees with μ_s on their common smooth subalgebra, and

$$T\rho_{\infty,b}(a) = \rho'_{\infty,b'}(\mu_\infty(a))T \quad \forall a \in \mathfrak{R}_{\infty,b}.$$

6. **one-particle data:** if i is a Hamiltonian label with lapse N and $\mu_C(i)$ has lapse N' , then for every q

$$\begin{aligned} UA_{D,N}(q) &= A'_{D,N'}(\kappa(q))U, \\ UA_{Y,N}(q) &= A'_{Y,N'}(\kappa(q))U, \\ U^\vee A_N(q)' &= A'_{N'}(\kappa(q))'U^\vee, \\ \widehat{\Lambda}_\pi^n(U^\vee)d\Gamma_n(A_N(q)') &= d\Gamma_n(A'_{N'}(\kappa(q))')\widehat{\Lambda}_\pi^n(U^\vee) \quad (n \geq 0). \end{aligned}$$

The last two equations follow from the first two, the declared dual, and projective functoriality, but are printed to fix their types.

7. **complete constraint family:** μ_C is a total bijection preserving constraint kind and transporting every field of the complete label record. For every $i \in \mathfrak{I}(\mathbf{Q})$,

$$T\widehat{C}_i = \widehat{C}'_{\mu_C(i)}T.$$

8. **global and clock status:** the topology, boundary, bundle, spin, matter-inventory, global-gauge, large-charge, anomaly, singular, clock domain, and clock-failure records commute with the displayed transport. Frozen numerical and source anchors match by identity.
9. **continuation:** the complete WDW-6 ledger is preserved entry by entry; no matching map may identify an open burden with an occupied result.

The solution bijection is deliberately absent from these hypotheses: it is a consequence of condition 7. Write $\mathbf{Q} \sim_5 \mathbf{Q}'$ when at least one complete candidate matching exists.

The matching data have explicit category operations. The identity uses the identity in every component. If $\mathbb{T} : \mathbf{Q} \rightarrow \mathbf{Q}'$, then $\mathbb{T}^{-1}$ uses

$$\iota_4^{-1}, \kappa^{-1}, \Gamma^{-1}, U_s^{-1}, U^{-1}, (U_s^\vee)^{-1}, (U^\vee)^{-1}, \mu_s^{-1}, \mu_\infty^{-1}, \mu_C^{-1}, \widetilde{T}^{-1}.$$

If $\mathbb{T}_1 : \mathbf{Q}_0 \rightarrow \mathbf{Q}_1$ and $\mathbb{T}_2 : \mathbf{Q}_1 \rightarrow \mathbf{Q}_2$, their composite uses the componentwise composites in that order, in particular

$$\mu_C^{21} = \mu_C^2 \circ \mu_C^1, \quad \mu_s^{21} = \mu_s^2 \circ \mu_s^1, \quad \mu_\infty^{21} = \mu_\infty^2 \circ \mu_\infty^1, \quad \widetilde{T}^{21} = \widetilde{T}^2 \circ \widetilde{T}^1.$$

Every compatibility equation is stable under these operations.

13.3 Solution and equivalence theorem

Theorem W5.12 (Formal joint-solution and proper-equivalence theorem).

The branch solution assignments are well-defined subspaces of the algebraic duals and are functorial under complete candidate matchings. They may be the zero subspace. The relation $\sim_5$ is an equivalence relation that preserves all nine standing-bearing compatibility classes, including the declared compact-open Bastiani topology, finite-degree topology, two separately typed representation levels, one-particle data, and the complete constraint-label family. It does not presuppose a measure, physical inner product, preferred quotient, or candidate nonemptiness beyond the zero functional.

Proof.

Theorem W5.11 gives $\widehat{C} : \Phi_5 \rightarrow \Phi_5$, so precomposition gives the typed transpose $\widehat{C}^\times : \Phi_5^\times \rightarrow \Phi_5^\times$. The kernel of a linear transpose is a vector subspace, and an arbitrary intersection of such kernels is a vector subspace. This proves both branch definitions, including the possibility that only the zero functional survives.

Let $\mathbb{T} : \mathbf{Q} \rightarrow \mathbf{Q}'$. If $\Psi' \in \text{Sol}_5(\mathbf{Q}')$, then for every $i \in \mathcal{I}(\mathbf{Q})$,

$$(T^\times \Psi')(\widehat{C}_i F) = \Psi'(T \widehat{C}_i F) = \Psi'(\widehat{C}'_{\mu_C(i)} T F) = 0.$$

Thus $T^\times$ maps the target joint annihilator into the source joint annihilator. Conversely, if $\Psi \in \text{Sol}_5(\mathbf{Q})$, then for $i' = \mu_C(i)$ and $F' \in \Phi_5'$,

$$((T^{-1})^\times \Psi)(\widehat{C}'_{i'} F') = \Psi(T^{-1} \widehat{C}'_{\mu_C(i)} F') = \Psi(\widehat{C}_i T^{-1} F') = 0.$$

Totality and surjectivity of μ_C cover every target constraint, so $(T^{-1})^\times$ is the inverse solution bijection. No solution bijection has been inserted as an independent premise.

The topology invoked by the comparison is not implicit. Section 12.1 defines $\tau_{b,\text{sec}}$ by compact derivative-test seminorms on each Bastiani mapping space and then takes the locally convex direct sum across finite exterior degrees. The wedge, redescription action, $\rho_{\infty,b}$, one-particle lifts, and regulated primitives are continuous there. Bicontinuity of $\widetilde{T}$ and its inverse therefore has a declared type and preserves the strong-dual, completed-degree, and mapping-space topologies.

The representation and one-particle squares are also domain-correct. The first representation square uses $\rho_{s,b}^{(0)}$ only on the initial seed subalgebra; the second uses $\rho_{\infty,b}$ on Φ_{5b} , which Theorem W5.11 proves invariant. The maps $U_s, U, U_s^\vee, U^\vee$ select the transported generators and make both algebra maps actual isomorphisms, not notation for an unspecified correspondence. The displayed Dirac/Yukawa square types the base point through κ ; its transpose and $d\Gamma$ squares follow on the strong dual and completed exterior degrees. There is no evaluation of a general strong-dual distribution on a nonsmooth fixed-rung spinor and no differentiation of a smooth-core section in a nonsmooth H^s direction.

The printed componentwise identity matching witnesses reflexivity. The componentwise inverse preserves all nine equations and witnesses symmetry. For transitivity, substitute the first square into the second; the result is the square for $\mathbb{T}_2 \circ \mathbb{T}_1$, with $\mu_s^2 \mu_s^1$, $\mu_\infty^2 \mu_\infty^1$, and $\mu_C^2 \mu_C^1$. Bijections, proper groupoid isomorphisms, filtered graded bicontinuous maps, and the one-particle squares are all closed under composition. Hence the complete matching tuples form a groupoid and $\sim_5$ is its object-equivalence relation. No Hilbert inner product or completion across particle degrees is used. $\square$

14. Candidate-family classification

14.1 Complete coordinate set

Within the construction class of this paper, the variable index is

$$\begin{aligned} \alpha_b = (b, \mathcal{U}_{5b}^s, \mathcal{G}_{5b}^{\text{prop}}, \mathcal{P}_b, \rho_b^{\text{pol}}, \rho_{s,b}^{(0)}, \rho_{\infty,b}, \mathcal{I}(\mathbf{Q}), \\ h_\star, \{L_{r,\star}\}_r, \chi_\epsilon, \epsilon, \nabla_b^{\text{conf}}, \mathfrak{o}_b, \mathfrak{o}_{\text{dens}}, \\ \mathfrak{C}_{5b}^{\text{ferm}}, \tau_{b,\text{sec}}, \mathcal{C}_{5b}^{\text{sec}}, \Phi_{5b}, \Phi_{5b}^\times, \mathcal{U}_{\text{clk}}, \text{ClockFail}). \end{aligned}$$

Here ρ_b^{pol} includes the bosonic configuration polarization and, on branch B , the graded holomorphic CAR polarization. The smoothing profile is the heat profile in the baseline family; replacing it is a new coordinate. The source $\mathfrak{H}_4^{\text{cl}}$, topology, bundle, spin structure, matter inventory, global gauge form, and numerical classical parameters are fixed anchors, not free entries of α_b . For branch B ,

$$\mathfrak{C}_{5B}^{\text{ferm}} = (\mathcal{S}_\infty, \mathfrak{f}_\infty, \{\mathfrak{F}_B^{(n)}\}_{n \geq 0}, \mathfrak{F}_{B,\text{fin}}, j_s, A_D, A_Y, \{d\Gamma_n\}_{n \geq 0}, \{\tau_{B,n}\}_{n \geq 0});$$

on branch A this coordinate is the trivial scalar degree-zero carrier.

The index sets are

$$A_5 = \{\alpha_A : \alpha_A \text{ meets all branch-A guards}\},$$

$$B_5 = \{\alpha_B : \alpha_B \text{ meets all branch-B guards}\}.$$

For each index, Theorems W5.2–W5.12 construct one $\mathbf{Q}_{5b, \alpha_b}$. This is a classification of the admitted WDW-5 candidate family. It is not a classification of every quantization of gravity.

14.2 Anchor, tensor/operator, and skin

A coordinate is an **anchor** when changing it changes the target, carrier class, boundary, topology, bundle, spin structure, global gauge form, or matter inventory. A coordinate is **tensor/operator content** when it changes the CCR/CAR action, regulated constraint, solution functor, clock status, anomaly status, or WDW-6 continuation. A difference is **skin** only when it is removed by a complete $\sim_5$ matching preserving every standing-bearing output.

Consequently, a smoothing profile, ordering, density map, field-space connection, probe completion, polarization, or fermionic completion is not skin merely because two formal equations look alike. Historical filenames, section ordering, dummy indices, or presentation changes are skin only when they alter none of the nine equivalence compatibility classes.

14.3 Classification theorem

Theorem W5.13 (Candidate-family classification theorem).

The sets A_5 and B_5 index every candidate admitted by the fixed construction class. Each formula in the constraint and solution package is determined by the fixed anchors together with the displayed variable coordinates. A change outside the fixed target is a target update; a change inside the target is standing-bearing unless a complete $\sim_5$ matching proves it to be skin. No equality of formal differential expressions alone establishes equivalence.

Proof.

The carrier and descent depend on $(b, \mathcal{U}, \mathcal{G}^{\text{prop}})$. The differential depends on the probe and chart data. The representation depends on the bosonic and fermionic polarizations. The regulated Hessian depends on $(h_*, L_*, \chi_\epsilon, \epsilon)$. The constraint action depends additionally on the ordering, density, and field-space connection ledgers. The smooth-core rigging, strong dual, completed exterior degrees, finite-particle filtration, compact-open Bastiani mapping-space topologies, locally convex direct-sum topology, two representation levels, primitive-stable ambient section algebra, common domain, and dual are constructed from those actions, and the solution functor depends on the full constraint-label family. The clock and continuation outputs retain their explicit source coordinates. Thus no formula contains an unlisted variable choice.

If a changed datum alters a fixed source or target coordinate, it leaves the declared theorem domain. Otherwise it changes one of the constructed standing-bearing objects unless a complete matching tuple supplies all nine compatibility classes, including the algebra maps, one-particle maps, and total constraint-label bijection. Theorem W5.12 proves that such a matching preserves the formal solution object and the attached ledgers. Hence the stated classification is complete within the admitted class. $\square$

15. QReg5 capstone and WDW-6 handoff

15.1 Occupied certificate

For a constructed candidate, define

$$\text{QReg5}(\mathbf{Q}_5) = \bigwedge_{r=1}^{16} Q5R_r.$$

The witnesses are:

Clause	Witness
Q5R-01	registered integrated WDW-4 identity and Theorem W5.1
Q5R-02	open metric-scalar chart, Theorem W5.2
Q5R-03	graded tetrad-SM chart, Theorem W5.2
Q5R-04	proper groupoid and local slice, Theorem W5.3
Q5R-05	tangent-dual density pairings, Theorem W5.4
Q5R-06	separating test algebras, Theorem W5.5
Q5R-07	independent Fréchet calculus, Theorem W5.5

Clause	Witness
Q5R-08	finite-rank type-jump exclusion, Theorem W5.6
Q5R-09	typed pivot-scale heat smoothing and regulated trace, Theorem W5.7
Q5R-10	CCR/CAR seeds, Theorem W5.8
Q5R-11	all linear candidates including six Lorentz directions, Theorem W5.9
Q5R-12	both Hamiltonians, including the complete degree-wise Dirac/Yukawa action, Theorem W5.10
Q5R-13	topologized completed exterior-section carrier, total primitives, two representation levels, and common domain, Theorem W5.11
Q5R-14	typed dual annihilator functors, Theorem W5.12
Q5R-15	typed compact-open carrier topology and composable matching package with domain-correct two-level representations, one-particle maps, and complete constraint-label bijection, Theorems W5.12–W5.13
Q5R-16	nonclaim and WDW-6 continuation ledgers, Theorem W5.14

15.2 Exported object

The exact WDW-6 input is

$$\mathfrak{H}_{5 \rightarrow 6} = (\mathfrak{H}_4^{\text{cl}}, \{\mathbf{Q}_{5A,\alpha}\}_{\alpha \in A_5}, \{\mathbf{Q}_{5B,\beta}\}_{\beta \in B_5}, \mathfrak{C}_{5B}^{\text{ferm}}, \text{QReg5}, \text{Sol}_5, \sim_5, \mathfrak{G}_6).$$

The continuation ledger $\mathfrak{G}_6$ contains:

1. a kinematical Hilbert completion or another declared topological carrier;
2. density of the algebraic core in that carrier;
3. one domain for constraints, adjoints, products, and commutators;
4. symmetry, closability, closures, and extension choices;
5. regulator removal, renormalization, or a controlled regulated endpoint;
6. comparison of factor orderings and smoothing coordinates;
7. quantum momentum, Gauss, and Lorentz closure;
8. Hamiltonian commutators and structure-function realization;
9. local, global, determinant-line, bordism, large-gauge, and large-diffeomorphism anomalies;
10. nontriviality of the joint kernel or rigged solution space;
11. a physical inner product or rigging map;
12. separating physical or relational observables;
13. compatibility of the local clock and its failure boundary with the surviving quantum object; and
14. semiclassical and source-attached covariant recovery.

Algebraic finite compositions on Φ_5 do not occupy rows 3, 5, 7, or 8: those rows require a topological operator domain, convergence, and exact quantum closure.

15.3 Capstone theorem

Theorem W5.14 (Configuration-space quantum candidate capstone).

On the exact regular compact classical branch exported by integrated WDW-4, there exist nonempty index sets A_5, B_5 and corresponding core and Standard-Model regulator-indexed algebraic candidate families such that every clause of QReg5 is occupied. The package $\mathfrak{H}_{5 \rightarrow 6}$ is source-faithful, branch-typed, target-indexed, and complete at proof-class level 2. It claims none of the WDW-6 survival conditions.

Proof.

The baseline choices used in Theorems W5.2–W5.11 provide at least one index in each of A_5 and B_5 : the selected Sobolev chart, total probe family, configuration/holomorphic polarization, heat regulator, flat chart connection, left-coefficient ordering, full $\mathfrak{spin}(1, 3)$ smearing, smooth-core/strong-dual rigging, degree-wise completed exterior carrier, compact-open Bastiani and finite-degree direct-sum topologies, separately typed fixed-rung and smooth-core representation levels, complete constraint-label records, section-valued Hessian action, and unguarded finite-primitive domain. The complete matching construction in Theorem W5.12 supplies explicit identity, inverse, and composition operations and derives the solution bijection from the total constraint-label square. Theorems

W5.1–W5.13 supply the witnesses Q5R-01 through Q5R-15 in the table. The complete nonclaim ledger is printed in Appendix D, and the fourteen-row continuation list above supplies Q5R-16.

The dependency is acyclic: the classical source precedes the configuration carriers; carriers precede differentials; differentials precede regulation and representation; those precede constraints; constraints precede the common domain; the common domain precedes the transpose solution functor; and the complete object precedes classification and handoff. AASC necessity governs the standing of each determinate object but substitutes for none of these positive constructions.

Therefore all sixteen certificate clauses hold. Since no Hilbert topology, limit, quantum commutator theorem, anomaly theorem, inner product, nontrivial-state theorem, or observable theorem appears in the proof, the conclusion stops exactly at the WDW-6 boundary. $\square$

16. Adversarial closure trace

The authoritative 42-row map is printed in Appendix F. This section records the load-bearing closure routes so that table status is never mistaken for proof.

16.1 Import and type-jump attacks

ADV5-01–ADV5-04 test whether the manuscript merely renames the ADM substitution, lets AASC perform quantization, argues backward from a candidate, or extends a finite-rank connection to configuration space. Theorems W5.1–W5.7 close these attacks constructively. The source is registered without backflow; the two Sobolev carriers and their Fréchet calculus are constructed independently; and the finite-rank theorem is used only inside spacetime coefficients.

16.2 Carrier, quotient, and differential attacks

ADV5-05–ADV5-13 test global linearization of the metric cone, a naive superspace quotient, erased improper transformations, nonseparating probes, false compact support, symbolic functional derivatives, unlicensed Riesz identifications, an undefined diagonal Hessian, and a hidden preferred background. Theorems W5.2–W5.7 close the chain. Positivity and coframe orientation are open local conditions; quotient descent is restricted to the accepted free regular slice; total probes separate the charts; derivatives land in continuous duals; and every Hessian trace follows the explicit $L^2 \rightarrow H^r \rightarrow H^{-r} \rightarrow H^{-r} \rightarrow L^2$ pivot chain. The kinetic coefficient gains no derivatives, while the two heat factors make the composition trace class. The reference metric and smoothing profile remain candidate coordinates.

16.3 Normalization, clock, and fermion attacks

ADV5-14–ADV5-20 test guessed DeWitt factors, hidden ordering, an external or preferred clock, deletion of the zero-body fermions, ignored second-class reduction, and a silent Fock measure. Theorems W5.4, W5.8, and W5.10 trace every coefficient to the accepted canonical forms. The clock is a constrained scalar with its local failure boundary. The full chiral exterior seed realizes $i\hbar$ times the reduced Dirac bracket and embeds injectively into the smooth-core strong-dual carrier. Its exterior powers are complete within each particle degree solely so the configuration- dependent Dirac/Yukawa action and target-valued heat trace are total. Completion across degrees, a Fock norm, determinant, Pfaffian, vacuum, and measure remain in $\mathfrak{G}_6$.

16.4 Constraint and common-domain attacks

ADV5-21–ADV5-28 test regression of the WDW-4B off-shell repair, omitted gauge or Lorentz sectors, unsmeared distributions, missing matter terms, an independent spin connection, circular domain construction, promotion to a dense Hilbert operator, and confusion about composition closure. Theorem W5.9 keeps the exact D plus G plus J convention and acts on every sector. Its Lorentz smearing retains all six $\mathfrak{spin}(1,3)$ directions, so $\iota_\xi \omega_{LC}(e)$ is in the declared class. Theorem W5.10 prints both complete Hamiltonians. Theorem W5.11 first proves every primitive total on $\mathcal{E}_{5b}^{\text{sec}}$, then constructs Φ_5 by unguarded finite-stage induction. On branch B , the strong-dual estimates prove that $q \mapsto A_N(q)'\eta$ is a smooth degree-one section without a finite-dimensional-range assumption; projective functoriality extends $d\Gamma_n(A_N(q)')$ to every completed degree; and the complete-target W5.7 estimate proves convergence of the section-valued Hessian trace in every target seminorm. These are the proof-level closures of ADV5-26 and ADV5-28. The theorem proves only algebraic single-action invariance and finite-word evaluation. Hilbert graph domains, convergence across regulators, adjoints, and commutator closure remain open.

16.5 Solution, anomaly, and equivalence attacks

ADV5-29–ADV5-38 test whether the dual annihilator is mislabeled a physical Hilbert space, whether nonzero solutions or quantum closure are assumed, whether local Standard-Model anomaly sums are overpromoted, whether formal resemblance replaces an intertwiner, whether regulator changes are called skin, whether the two branches are conflated or disconnected from WDW-4, and whether a kinematical inner product is treated as physical. Theorems W5.12–W5.14 close these attacks. The transpose equations are typed, the joint subspace may be zero, all anomaly rows remain in WDW-6, and $\sim_5$ preserves nine explicit coordinates. In particular, ADV5-33 is closed by two independent type checks. First, the ambient mapping spaces carry the printed compact-open Bastiani seminorm topology and their finite-particle sum carries the locally convex direct-sum topology, so a bicontinuous graded intertwiner is a defined object. Second, the full fixed-rung representation $\rho_{s,b}^{(0)}$ is intertwined only on the initial seed subalgebra, while the smooth-core representation $\rho_{\infty,b}$ is intertwined on its W5.11-invariant domain Φ_{5b} . No nonsmooth fixed-rung direction is made to act on the inverse-limit section carrier. Third, the comparison witness contains actual isomorphisms μ_s, μ_∞ , compatible primal/dual one-particle maps, a base and proper-groupoid transport, and a total bijection μ_C of complete constraint labels. Every representation and one-particle square uses those maps, and the solution bijection is derived from the surjective constraint square rather than assumed.

16.6 Scope and proof-status attacks

ADV5-39–ADV5-42 test overclassification of topology and bundles, derivation of numerical parameters, ledger-only proof, and theft of WDW-6 burdens. The fixed target ledger, fourteen theorem proofs, sixteen certificate witnesses, exact nonclaim ledger, and exported continuation object close them. In particular, ADV5-41 is witnessed by the pivot proof, six-direction Lorentz proof, strong-transpose estimate, degree-wise $d\Gamma$ extension, complete-target trace estimate, and only then the primitive-totality ledger. The comparison witnesses are likewise the compact derivative-test seminorm construction, continuity proof, and two-level seed-action squares before their audit rows are marked closed. The final witness is the complete matching tuple: all of its maps are typed, its identity, inverse, and composition are componentwise, and its constraint-label bijection is used in both directions in the W5.12 annihilator proof. Thus neither a correspondence symbol nor a ledger status substitutes for a map. Every pass status is downstream of theorem-bearing text.

17. Exact endpoint

The result is stronger than the historical formal Wheeler-DeWitt expression in four precise ways.

First, the configuration variables inhabit actual local Sobolev carriers. Second, the functional derivatives are continuous Fréchet differentials, with every coincident second-order contraction replaced by a declared, pivot-typed positive-regulator trace. Third, the complete linear and Hamiltonian constraint families, including all six Lorentz directions, act on one explicitly generated algebraic domain after their primitives are proved total on a concrete ambient section algebra. The Standard-Model carrier admits configuration-dependent exterior sections, continuous Dirac/Yukawa $d\Gamma$ actions on each completed degree, and seminorm-convergent section-valued Hessian contractions; the common domain therefore has a typed transpose on its algebraic dual. Fourth, the core and Standard-Model families retain their full classical source, target, clock, anomaly, and continuation coordinates. Their proper-equivalence relation is witnessed by composable transport packages, including the two representation-algebra maps, primal/dual one-particle maps, and a bijection of complete constraint records.

The result is weaker than a quantum theory in equally precise ways. It provides no kinematical measure, Hilbert completion, dense operator core, adjoint, closability result, regulator limit, renormalized constraint, Hamiltonian commutator theorem, anomaly theorem, nonzero solution, physical inner product, rigging map, or separating observable algebra. These absences are not qualifications added after the proof; they determine its type.

18. Conclusion

Integrated WDW-4 delivered a nonempty source-attached classical constraint interior with a scalar/clock core, a torsion-free Standard-Model extension, an exact nonconflating lift, and a complete quantum-scope boundary. WDW-5 has now crossed the next type boundary.

The metric-scalar and tetrad-Standard-Model configuration carriers are constructed as distinct local objects. Proper redescription acts before descent. Their first and second variations arise from their own Sobolev smooth structures, not from an untyped extension of the spacetime connection. Heat smoothing makes the correctly pivot-typed kinetic Hessian traces finite at every positive regulator. The canonical forms fix the CCR and CAR coefficients, and the complete Lorentz action retains boosts and rotations. The smooth-core strong dual and its degree-wise completed exterior powers carry the complete configuration-dependent Dirac/Yukawa action

without a fixed-monomial restriction. Every primitive is total on the Bastiani-smooth finite-particle section algebra before the exact linear and Hamiltonian constraints preserve one unguardedly constructed algebraic domain. Its transpose gives a well-typed formal joint-annihilator problem. Candidate comparison preserves the carrier topology, degree filtration, and every other standing-bearing coordinate rather than relying on equal-looking notation. Its bicontinuity is evaluated in the declared compact-open Bastiani/direct-sum topology, and its CCR/CAR conditions distinguish the full fixed-rung action on the initial seed algebra from the smooth-core action on the generated domain. The comparison witness explicitly transports the base, proper groupoid, fixed-rung and smooth-core algebras, Dirac/Yukawa data, and every complete constraint label. These maps have componentwise identity, inverse, and composition, and the formal-solution bijection follows from their constraint square.

This completes Configuration-Space Quantum Constraint Candidate Closure. The surviving object is the full family

$$\mathfrak{H}_{5 \rightarrow 6} = (\mathfrak{H}_4^{\text{cl}}, \{\mathbf{Q}_{5A,\alpha}\}_{\alpha \in A_5}, \{\mathbf{Q}_{5B,\beta}\}_{\beta \in B_5}, \mathfrak{C}_{5B}^{\text{ferm}}, \text{QReg5}, \text{Sol}_5, \sim_5, \mathfrak{G}_6).$$

WDW-6 receives this family without a hidden promotion. Its task is to decide whether any candidate survives Hilbert-domain, regulator, ordering, commutator, anomaly, state, inner-product, observable, clock, and covariant recovery tests.

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Appendix A. Theorem dependency and certificate ledgers

A.1 Theorem dependency ledger

ID	Theorem target	Depends on	Status	Downstream use
W5.1	Exact classical import and no-backflow theorem	Integrated WDW-4 publication edition; Primitive Quantization Obstruction	PROVED	W5.2-W5.14
W5.2	Sobolev configuration-carrier theorem	W5.1; Ebin/Sobolev manifold imports	PROVED	W5.3-W5.14
W5.3	Proper-redescription groupoid descent theorem	W5.1-W5.2; accepted WDW-2 and WDW-4 groupoids	PROVED	W5.5; W5.9; W5.12-W5.14
W5.4	Configuration tangent/cotangent and symplectic pairing theorem	W5.1-W5.2; WDW-1/4 canonical conventions	PROVED	W5.5; W5.7-W5.11
W5.5	Cylinder/local differential theorem	W5.2-W5.4	PROVED	W5.7-W5.14
W5.6	Finite-rank nonextension and independent differential theorem	W5.1; W5.4-W5.5; covariant-differentiation boundary	PROVED	W5.7-W5.14
W5.7	Pivot-typed regulator and field-space contraction theorem	W5.4-W5.6; elliptic smoothing import; trace-ideal estimate	PROVED	W5.10-W5.14
W5.8	CCR/CAR representation-seed theorem	W5.4-W5.5; WDW-4B Dirac bracket	PROVED	W5.9-W5.14
W5.9	Full six-direction linear quantum-constraint candidate theorem	W5.3-W5.5; W5.8; accepted classical moment maps	PROVED	W5.11-W5.14
W5.10	Regulator-indexed Hamiltonian candidate theorem	W5.4-W5.8; accepted classical Hamiltonians; smooth-core Dirac Yukawa scale estimates; degree-wise dGamma; complete-target trace	PROVED	W5.11-W5.14
W5.11	Primitive-total common algebraic action-domain theorem	W5.5; W5.7-W5.10; compact-open topologized strong-dual completed exterior-section algebra; all primitive continuity and totality lemmas; two representation levels	PROVED	W5.12-W5.14
W5.12	Formal joint-solution and proper-equivalence theorem	W5.3; W5.9-W5.11; compact-open and finite-degree topologies; explicit base groupoid representation one-particle and total constraint-label matching maps; componentwise identity inverse composition	PROVED	W5.13-W5.14
W5.13	Candidate-family classification theorem	W5.1-W5.12; fermionic rigging topology filtration two representation levels complete label family and composable matching package	PROVED	W5.14; WDW-6
W5.14	Configuration-space quantum candidate capstone	W5.1-W5.13; QReg5; fermionic section-module and repaired equivalence-interface ledgers	PROVED	WDW-6

A.2 QReg5 certificate

ID	Clause	Witness	Status	Gate
Q5R-01	Exact integrated WDW-4 source identity and complete handoff retained	Publication handoff declaration	PROVED	W5.1
Q5R-02	Core metric-scalar Sobolev configuration chart constructed	Open Hilbert-manifold chart around z_0	PROVED	W5.2
Q5R-03	Tetrad-spin-gauge-Higgs-fermion graded chart constructed	Even chart plus odd chiral test carrier around w_{delta}	PROVED	W5.2

ID	Clause	Witness	Status	Gate
Q5R-04	Proper-redescription groupoid action and local regular descent constructed	Equivariant test algebra and accepted local slice	PROVED	W5.3
Q5R-05	Continuous tangent/cotangent and canonical pairings typed	Sobolev duals and inherited symplectic normalization	PROVED	W5.4
Q5R-06	Separating cylinder/local test algebras constructed	Probe separation and chart support proof	PROVED	W5.5
Q5R-07	First and second variational differentials independently proved	Frechet formulas and continuity estimates	PROVED	W5.5-W5.6
Q5R-08	Finite-rank spacetime-to-configuration type jump excluded	Explicit nonextension theorem and independent construction ledger	PROVED	W5.6
Q5R-09	Smoothing and regulator family fully typed	L2-Hr-H-r-H-r-L2 pivot; heat maps; order-zero kinetic multiplier; trace-ideal estimate	PROVED	W5.7
Q5R-10	CCR and graded CAR representation seeds constructed	Smear algebraic commutator and anticommutator checks	PROVED	W5.8
Q5R-11	All linear constraint candidates constructed	Spatial; internal Gauss; all six spin(1,3) Lorentz generators; exact improved convention	PROVED	W5.9
Q5R-12	Core and SM regulator-indexed Hamiltonian candidates constructed	Complete gravitational and matter terms plus smooth strong-transpose Dirac Yukawa dGamma action and section-valued trace	PROVED	W5.10
Q5R-13	One common algebraic action domain constructed	Compact-open Bastiani and finite-degree topologies; completed exterior-section carrier; every primitive continuous and total; two representation levels; unguarded finite-word domain; typed transpose	PROVED	W5.11
Q5R-14	Formal joint-solution functors defined without nonemptiness promotion	Branch-indexed annihilator conditions in Phi dual after completed-carrier totality	PROVED	W5.12
Q5R-15	Candidate equivalence and standing classifier constructed	Bicontinuous filtered graded matching with explicit base groupoid $\mu_s \mu_\infty$ primal-dual one-particle and total bijective constraint-label maps; componentwise identity inverse composition; solution bijection derived from the constraint square	PROVED	W5.12-W5.13
Q5R-16	Complete WDW-6 continuation and nonclaim ledgers attached	Survival-test handoff with no quantum-closure promotion	PROVED	W5.14

Appendix B. Frozen target, imports, and proof class

B.1 Proof-class ladder

Level	ID	Object	WDW-5 status	Evidence
1	PC5-01	Typed formal functional differential expression	IN_SCOPE	Every symbol has carrier pairing smearing density and branch

Level	ID	Object	WDW-5 status	Evidence
2	PC5-02	Regulator-indexed algebraic operator family on common test domain	IN_SCOPE	Degreewise completed exterior-section carrier with compact-open Bastiani and finite-degree direct-sum topologies; separately typed fixed-rung and smooth-core representations; smooth strong-transpose and dGamma actions; target-valued trace; every primitive continuous and total before unguarded finite-word closure; all constraints preserve the domain and transpose to its algebraic dual; proper equivalence uses explicit composable matching maps and a total constraint-label bijection
3	PC5-03	Densely defined operator on a kinematical Hilbert space	OPEN_WDW6	Measure or Hilbert completion and density proof
4	PC5-04	Invariant closable or closed common operator domain	OPEN_WDW6	Composition adjoint and graph-closure analysis
5	PC5-05	Regulator removal or controlled regulated endpoint	OPEN_WDW6	Convergence renormalization and independence proof
6	PC5-06	Anomaly-free joint quantum representation	OPEN_WDW6	Exact commutator closure at frozen target
7	PC5-07	Nontrivial physical state space and physical inner product	OPEN_WDW6	Rigging or equivalent construction and positivity
8	PC5-08	Separating physical or relational observables	OPEN_WDW6	Observable action and separation theorem

B.2 Target-freeze ledger

ID	Coordinate	Frozen value	Class	Guard
T5-01	Spatial manifold	Compact connected boundaryless hyperbolic Sigma	ANCHOR	Topology change is a target update
T5-02	Orientation	Accepted orientation	ANCHOR	Orientation reversal is separately typed
T5-03	Spin structure	One accepted spin structure	ANCHOR	No sum over spin structures
T5-04	Boundary package	Empty compact boundary package	ANCHOR	No edge or asymptotic sector
T5-05	Regularity	Regular Sobolev chart with s greater than $7/2$	ANCHOR	Singular strata retained outside chart
T5-06	Core gravity carrier	Riemannian metric chart	ANCHOR	No connection-variable substitution by notation
T5-07	SM gravity carrier	Oriented nondegenerate tetrad chart with derived torsion-free spin connection	ANCHOR	No independent affine or spin connection
T5-08	Clock sector	Accepted real scalar local clock and failure boundary	TENSOR	No global or preferred time
T5-09	Internal gauge global form	Declared Standard Model Z6 quotient branch	ANCHOR	Not derived from local representation data
T5-10	Gauge bundle	Accepted trivial principal bundle	ANCHOR	Nontrivial bundles are target variants
T5-11	Line sector	No line insertion	ANCHOR	Line spectrum remains global-form data
T5-12	Fermion content	Three-generation minimal chiral Standard Model without ν_R	ANCHOR	BSM fermions require target update
T5-13	Higgs content	One complex doublet of hypercharge $1/2$	ANCHOR	No extra scalar branch
T5-14	Classical parameters	Inherited fixed couplings Yukawas Higgs potential theta and cosmological convention	TENSOR	No numerical derivation
T5-15	Configuration polarization	Metric-scalar core and tetrad-SM extension	CANDIDATE	Alternatives require target update or proved equivalence
T5-16	Configuration smooth structure	Sobolev-Hilbert manifold charts	CANDIDATE	No formal superspace without topology

ID	Coordinate	Frozen value	Class	Guard
T5-17	Descent interface	Proper-redescription groupoid plus local regular slice	TENSOR	No naive global orbit manifold
T5-18	Test carrier	Smooth separating cylinder/local algebra	CANDIDATE	Probe family must separate chart
T5-19	Fermionic quantum carrier	Topologized degreewise completed CAR/exterior section module with finite particle support and explicit primal-dual matching maps	CANDIDATE	No hidden Fock completion or nonsmooth full-carrier contraction
T5-20	Regulator family	Declared elliptic smoothing kernels K_{ϵ}	CANDIDATE	Reference metric and profile are recorded
T5-21	Ordering	Explicit finite/infinite parameter ledger $\mathfrak{O}$	CANDIDATE	No preferred ordering claim
T5-22	Common action domain	Generated algebraic test module inside compact-open Bastiani section carrier with smooth-core action declared dual and total complete-label matching	TENSOR	No Hilbert-domain or full fixed-rung action promotion
T5-23	Solution object	Formal joint annihilator in algebraic dual	TENSOR	May be zero and is not H_{phys}
T5-24	Quantum closure status	Open WDW-6 ledger	OBSTRUCTION	No anomaly inner-product or observable closure in WDW-5

B.3 Import ledger

ID	Source	Payload	Use	Forbidden promotion	Status
IMP5-01	Integrated WDW-4 publication project	Exact 4A and 4B theorem chains plus typed interface and W4Reg	Controlling classical source	No modification of accepted classical claims	VERIFIED
IMP5-02	Integrated WDW-4 handoff declaration	Published endpoint and authorized WDW-5 handoff	Registration authority	Not a quantum verdict	VERIFIED
IMP5-03	WDW-1	Canonical variables symplectic normalization Hamiltonian densities and boundaryless convention	CCR and Hamiltonian normalization	No quantum representation supplied	INHERITED
IMP5-04	WDW-2	Proper-redescription groupoid and faithful quotient discipline	Configuration groupoid action	No globally smooth superspace supplied	INHERITED
IMP5-05	WDW-3	Hypersurface-deformation and refoliation theorem	Classical structure-function target	No quantum commutator closure supplied	INHERITED
IMP5-06	WDW-4A	Scalar regular chart local clock and exact constraint system	Core candidate branch	No functional derivative or quantum clock supplied	INHERITED
IMP5-07	WDW-4B	Tetrad spin gauge Higgs fermion Dirac bracket constraints and anomaly ledger	SM candidate branch	No CAR Hilbert representation or quantum anomaly theorem supplied	INHERITED
IMP5-08	Primitive Quantization Obstruction	No primitive map from classical gravity to a quantum object	No-backflow guard	Does not prove candidate existence	DECLARED
IMP5-09	Covariant differentiation theorem	Finite-rank transport-before-differentiation and Levi-Civita branch	Method and spacetime input only	No superspace derivative supplied	DECLARED
IMP5-10	Ebin manifold and slice framework	Smooth manifold of metrics and regular slice method	Metric configuration chart and descent hypotheses	No quantum measure or Hamiltonian operator supplied	EXTERNAL
IMP5-11	DeWitt canonical quantum gravity	Classical supermetric and historical formal constraint expression	Comparison and convention audit	Formal expression is not a defined operator	EXTERNAL

ID	Source	Payload	Use	Forbidden promotion	Status
IMP5-12	Ashtekar-Lewandowski projective calculus	Connection-space cylinder and projective comparison architecture	Comparison branch for gauge or connection alternatives	Does not prove metric/tetrad equivalence	EXTERNAL
IMP5-13	Algebraic CCR/CAR theory	Smear canonical representation seeds	Test-algebra construction	No uniqueness or Hilbert completion without hypotheses	EXTERNAL
IMP5-14	Giulini-Marolf refined algebraic quantization	Rigging-map comparison and uniqueness conditions	WDW-6 continuation anchor	No gravity group-averaging theorem imported	EXTERNAL
IMP5-15	Schrodinger Dynamics as the Unique Minimal First-Response Hamiltonian Projection of AASC Quantum Standing	Downstream projective-Hilbert and Hamiltonian role comparison	Comparison only; no WDW-5 premise or construction	Scalar flat-space Hamiltonian exhaustion is not transplanted to gravitational configuration space	COMPARISON
IMP5-16	Admissibility and Constraint-Induced Rigidity of Physical Quotient Structure	Regular quotient fidelity no-second-classifier and observable-completeness discipline	Groupoid descent and candidate-equivalence boundary	No quantization functor from ADM phase space to Hilbert space is supplied	VERIFIED
IMP5-17	Structural Unification of Gravity and Quantum Dynamics under AASC	Joint-ledger and canonical Wheeler-DeWitt route audit	Source target record and continuation discipline	No constraint closure physical state or quantum admissibility theorem is supplied	VERIFIED

Appendix C. Differential and operator ledgers

C.1 Configuration differential ledger

ID	Object	Domain	Codomain	Construction	Status	Guard
D5-01	Configuration chart U_{5A}	$\widehat{Riem}$'s times $\widehat{H}$'s	Sobolev Hilbert manifold	Open cone and product chart	PROVED	Regular core locus only
D5-02	Configuration chart U_{5B}	Coframe gauge Higgs spinor clock product	Graded local carrier	Even Sobolev chart plus odd test module	PROVED	Fixed bundle and spin sector
D5-03	First differential dF	Smooth test functional	Continuous tangent dual	Frechet derivative	PROVED	No pointwise density identification without pairing
D5-04	Second differential d2F	Smooth test functional	Continuous bilinear forms	Second Frechet derivative	PROVED	No coincident contraction without regulator
D5-05	Cylinder probe family	Configuration chart	Finite coordinate tuples	Separating continuous linear probes	PROVED	Must distinguish all declared chart directions
D5-06	Fermionic derivatives and one-particle lifts	Embedded CAR seed and completed exterior degrees	Finite-degree strong-dual section module	Left/right seed derivatives plus smooth transpose and degreewise dGamma	PROVED	Signs fixed by WDW-4B Dirac bracket and full-carrier contractions use smooth spinors
D5-07	Smoothing kernel K_{ϵ}	Sobolev dual fields	Smooth fields	Elliptic functional calculus of $\Delta_{\star}$	PROVED	Reference metric and profile recorded
D5-08	Regulated Hessian contraction	L_2 to H_r to H -r to L_2 pivot	Scalar trace or complete-target exterior-degree section	Two Hilbert-Schmidt legs bounded Hessian order-zero kinetic multiplier and target-seminorm convergence	PROVED	Defined only for epsilon greater than zero; no H -r to H_r multiplier
D5-09	Field-space fundamental vector fields	Proper groupoid Sobolev scale	Smooth-core tangent with one finite-rung loss	Derivative of accepted action	PROVED	Singular and improper directions guarded
D5-10	Local slice descent	Invariant or equivariant test algebra	Regular quotient chart	Accepted slice plus restriction	PROVED	No global superspace manifold claim

C.2 Fermionic section-module ledger

ID	Object	Domain	Codomain	Construction	Status	Guard
F5S-01	Smooth spinor core	Intersection of all H^s spinor rungs	Nuclear Frechet space S_∞	Projective Sobolev inverse limit	PROVED	No fixed-rung same-order claim for Dirac action
F5S-02	Strong-dual fermionic carrier	S_∞	Complete nuclear DF space f_∞	Strong dual and inductive Sobolev-dual limit	PROVED	No Fock norm or fermionic measure
F5S-03	Fixed-rung CAR embedding	Dual of H^s test space	f_∞ and constant exterior sections	Dense-core restriction j_s and algebraic exterior functor	PROVED	Full-carrier contractions use smooth test spinors
F5S-04	Completed exterior degrees	f_∞	Completed projective alternating power F_B^n	Projective tensor completion followed by anti-symmetrization	PROVED	Completion is degreewise only
F5S-05	Finite-particle section algebra	Smooth branch-B configuration core	Locally convex direct sum of compact-open Bastiani F_B^n -valued section spaces	Compact derivative-test seminorms and continuous graded wedge product	PROVED	No completion across particle number
F5S-06	Dirac and Yukawa one-particle maps	S_∞	S_∞	Rungwise differential and multiplier estimates with all configuration derivatives	PROVED	Derived torsion-free spin connection only
F5S-07	Smooth strong transpose	f_∞	f_∞	Bounded-set seminorm estimates for the strong transpose	PROVED	No unproved fixed Sobolev-dual preservation
F5S-08	Exterior dGamma action	F_B^n	F_B^n	Continuous projective tensor lift and antisymmetrization	PROVED	Ordering degree shifts must be separately typed
F5S-09	Section-valued heat trace	Smooth F_B^n -valued sections	Smooth F_B^n -valued sections	Target-seminorm estimate with two Hilbert-Schmidt legs and completeness	PROVED	Positive regulator only
F5S-10	Complete primitive continuity and totality	Topologized E_5B sec	Topologized E_5B sec	Dirac Yukawa dGamma smooth-core representation ordering pieces and Hessian proved continuous and total before domain recursion	PROVED	No ledger-only where-defined or full fixed-rung action closure

C.3 Equivalence-interface ledger

ID	Object	Domain	Codomain	Construction	Status	Guard
E5I-01	Degree-wise section topology	Smooth maps from the inverse-limit configuration core into one exterior degree	Hausdorff locally convex mapping space	Compact-open Bastiani derivative seminorms	PROVED	No Hilbert or operator graph topology
E5I-02	Finite-particle topology	Algebraic finite-degree section sum	Locally convex direct sum τ_b sec	Direct sum of the degree-wise compact-open topologies	PROVED	No completion across particle degree
E5I-03	Continuous ambient algebra and primitives	Topologized ambient section carrier	Topologized ambient section carrier	Leibniz compact-set estimates projective exterior continuity and uniform regulated trace estimates	PROVED	Continuity is not closability or Hilbert boundedness
E5I-04	Full fixed-rung representation	Fixed-rung CCR CAR seed and initial test algebra	Initial test algebra	Representation $\rho_{s_b_0}$ from W5.8	PROVED	No extension of nonsmooth directions to the inverse-limit carrier

ID	Object	Domain	Codomain	Construction	Status	Guard
E5I-05	Smooth-core representation	Smooth-core CCR CAR algebra and ambient section carrier	Continuous ambient endomorphisms and invariant Φ_5	Configuration multiplication smooth momentum derivatives exterior multiplication and smooth contraction	PROVED	No evaluation of a strong-dual distribution on a nonsmooth spinor
E5I-06	Representation-algebra matching package	Fixed-rung and smooth-core algebras of one candidate	Corresponding algebras of the target candidate	Explicit unital graded isomorphisms μ_s and μ_∞ induced by typed base and primal-dual generator transport; T_0 and T squares use those maps on their actual domains	PROVED	No correspondence symbol and no fixed-rung equation on all of Φ_5
E5I-07	One-particle and constraint matching package	Smooth one-particle data and complete constraint-label set of one candidate	Corresponding one-particle data and complete label set of the target candidate	Compatible U_s U_{s_dual} U_dual maps with Dirac Yukawa transpose and $d\Gamma$ squares plus a total bijection μ_C preserving every operative label field	PROVED	Solution bijection is derived from the surjective constraint square
E5I-08	Composable topological candidate equivalence	Two complete ambient candidate carriers	Equivalence groupoid and derived bijection of formal solution functors	Bicontinuous filtered graded ambient isomorphism with explicit source base groupoid representation one-particle and constraint maps; identity inverse and composition are componentwise	PROVED	No equivalence from formal resemblance or undeclared matching data

C.4 Operator-candidate ledger

ID	Branch	Candidate	Index	Domain	Output	Status	WDW-6 burden
O5-01	Core	Metric configuration multiplication	Test tensor	Φ_5A	Φ_5A	PROVED	Hilbert boundedness
O5-02	Core	Metric momentum derivation	Test tensor	Φ_5A	Φ_5A	PROVED	Adjoint and closure
O5-03	Core	Scalar configuration multiplication	Test scalar	Φ_5A	Φ_5A	PROVED	Hilbert boundedness
O5-04	Core	Scalar momentum derivation	Test scalar	Φ_5A	Φ_5A	PROVED	Adjoint and closure
O5-05	Both	Improved spatial generator	Shift ξ	Φ_5	Φ_5	PROVED	Exponent and anomaly
O5-06	SM	Internal Gauss generator	Gauge parameter λ	Φ_5B	Φ_5B	PROVED	Large gauge anomaly
O5-07	SM	Full local Lorentz generator	Six-component $\text{spin}(1,3)$ parameter ϑ_{IJ}	Φ_5B	Φ_5B	PROVED	Spin or determinant anomaly
O5-08	Core	Regulated gravitational kinetic term	Lapse N epsilon and pivot rung	Φ_5A	Φ_5A	PROVED	Limit and ordering
O5-09	Core	Gravity potential multiplication	Lapse N	Φ_5A	Φ_5A	PROVED	Domain and symmetry

ID	Branch	Candidate	Index	Domain	Output	Status	WDW-6 burden
O5-10	Core	Scalar-clock Hamiltonian term	Lapse N and epsilon	Phi 5A	Phi 5A	PROVED	Clock and domain
O5-11	SM	Gauge Hamiltonian term	Lapse N and epsilon	Phi 5B	Phi 5B	PROVED	RG and anomaly
O5-12	SM	Higgs Hamiltonian term	Lapse N and epsilon	Phi 5B	Phi 5B	PROVED	RG and domain
O5-13	SM	Fermionic Hamiltonian term	Lapse N and explicit fermionic ordering	Completed-degree exterior sections	Phi 5B	PROVED	Fock completion CAR determinant anomaly and vacuum
O5-14	Both	Full Hamiltonian constraint candidate	Lapse N epsilon pivot and ordering	Phi 5	Phi 5	PROVED	Algebra regulator limit and closed domain
O5-15	Both	Formal joint solution functor	All allowed smearings	Algebraic dual of completed-carrier Phi 5	Subspace or zero	PROVED	Kernel nontriviality and physical inner product
O5-16	Both	Complete candidate matching tuple	Two complete candidates	Topologized ambient carriers both representation domains one-particle data and complete constraint-label sets	Derived solution bijection carrier filtration and ledger comparison	PROVED	Equivalence or inequivalence theorem

Appendix D. Exact nonclaims

D.1 Nonclaim ledger

ID	Nonclaim
NC5-01	WDW-5 does not construct a primitive quantization map from general relativity
NC5-02	WDW-5 does not derive the AASC kernel from a successful quantum candidate
NC5-03	The finite-rank spacetime connection theorem does not define a superspace functional derivative
NC5-04	The Sobolev configuration chart is not asserted to be the unique quantum configuration carrier
NC5-05	The local regular quotient is not asserted to be a globally smooth superspace
NC5-06	The formal notation δ by δh is not used as an unregulated coincident operator
NC5-07	No canonical translation-invariant or diffeomorphism-invariant measure on metric configuration space is claimed
NC5-08	No preferred smoothing kernel reference metric cutoff profile or factor ordering is claimed
NC5-09	No Hilbert completion density common invariant domain adjoint or closability theorem is claimed
NC5-10	No regulator removal renormalized limit or regulator independence theorem is claimed
NC5-11	No quantum hypersurface-deformation or Hamiltonian-constraint commutator closure is claimed
NC5-12	No local global determinant-line bordism large-gauge or dynamical-gravity anomaly closure is claimed
NC5-13	No nonzero formal joint solution is claimed
NC5-14	The formal joint solution functor is not called the physical Hilbert space
NC5-15	No physical inner product rigging map group averaging or master-constraint completion is claimed
NC5-16	No separating physical or relational observable algebra is claimed
NC5-17	The scalar clock is not promoted to global preferred or quantum time
NC5-18	The core metric-scalar and SM tetrad-spin-gauge candidates are not asserted equivalent
NC5-19	No Fock vacuum fermionic measure determinant Pfaffian or CAR uniqueness theorem is claimed
NC5-20	No nontrivial topology bundle boundary edge asymptotic line or singular-stratum sector is classified
NC5-21	No connection-variable loop BRST reduced or path-integral presentation is declared equivalent by analogy

ID	Nonclaim
NC5-22	No numerical coupling Yukawa Higgs theta cosmological or Planck-scale parameter is derived

Appendix E. Acceptance conditions

E.1 Acceptance-condition adjudication

ID	Condition	Proof gate	Adjudication
AC5-01	Integrated WDW-4 source edition is identified	W5.1	PASS
AC5-02	Integrated WDW-4 theorem object matches the cited edition	W5.1	PASS
AC5-03	Integrated WDW-4 publication handoff is explicit	W5.1	PASS
AC5-04	Complete integrated handoff tuple is reproduced without dropped coordinate	W5.1	PASS
AC5-05	Coupled covariant-history class remains distinct from canonical datum and lift	W5.1	PASS
AC5-06	Fixed-target and open-quantum ledgers are imported in full	W5.1	PASS
AC5-07	WDW-4A and WDW-4B phase carriers and constraint maps remain distinct	W5.1	PASS
AC5-08	No downstream quantum claim is used to repair an imported classical theorem	W5.1	PASS
AC5-09	Compact connected boundaryless oriented spin topology is unchanged	W5.1-W5.2	PASS
AC5-10	Sobolev threshold and every staggered field regularity are explicit	W5.2	PASS
AC5-11	Core metric-scalar configuration chart is an actual open Hilbert-manifold chart	W5.2	PASS
AC5-12	SM coframe-gauge-Higgs-spinor-clock carrier is fully typed	W5.2	PASS
AC5-13	Even and odd directions are distinguished in the graded chart	W5.2	PASS
AC5-14	Nondegenerate oriented coframe condition is open on the selected chart	W5.2	PASS
AC5-15	Fixed bundle global form spin structure and matter inventory are retained	W5.2	PASS
AC5-16	Proper spatial action is imported with its exact regularity and scope	W5.3	PASS
AC5-17	Internal gauge and Lorentz actions are present on the SM chart	W5.3	PASS
AC5-18	Local slice descent is restricted to the accepted regular free locus	W5.3	PASS
AC5-19	Singular stabilizers and nonproper transformations are retained as guards	W5.3	PASS
AC5-20	No naive global superspace manifold is assumed	W5.3	PASS
AC5-21	Tangent spaces and continuous duals are declared sector by sector	W5.4	PASS
AC5-22	Density pairings and index placements are explicit	W5.4	PASS
AC5-23	Canonical normalization is imported exactly from WDW-1 and WDW-4	W5.4	PASS
AC5-24	Fermionic left-right derivative and sign conventions follow the reduced Dirac bracket	W5.4	PASS
AC5-25	Probe families are continuous and separate every declared chart direction	W5.5	PASS
AC5-26	Cylinder/local test algebras are defined without compact-support fiction on infinite-dimensional charts	W5.5	PASS
AC5-27	First Frechet differential formula is proved	W5.5	PASS
AC5-28	Second Frechet differential formula is proved	W5.5	PASS
AC5-29	Differential continuity estimates are stated in the selected topology	W5.5	PASS
AC5-30	Graded derivatives are constructed on the algebraic fermion module	W5.5	PASS
AC5-31	Spacetime finite-rank connection and configuration-space differential are explicitly separated	W5.6	PASS
AC5-32	Every configuration differential fact used later is independently proved or exactly imported	W5.6	PASS
AC5-33	Identity-principal-symbol language is not silently transported to superspace	W5.6	PASS
AC5-34	Reference metric and elliptic operators defining K_{ϵ} are explicit	W5.7	PASS

ID	Condition	Proof gate	Adjudication
AC5-35	Smoothing profile epsilon range and bundle transport are explicit	W5.7	PASS
AC5-36	Adjoint and density conversion conventions for the kernel are explicit	W5.7	PASS
AC5-37	Regulated Hessian contraction follows the explicit L2-Hr-H-r-H-r-L2 pivot and is trace class for every positive epsilon	W5.7	PASS
AC5-38	No unregulated coincident second derivative is called an operator	W5.7	PASS
AC5-39	Reference-background and profile dependence are recorded as candidate coordinates	W5.7	PASS
AC5-40	Core configuration and momentum seeds satisfy the exact smeared CCR	W5.8	PASS
AC5-41	Scalar-clock CCR seed uses the inherited normalization	W5.8	PASS
AC5-42	SM graded seed satisfies the CAR determined by the Dirac-reduced bracket	W5.8	PASS
AC5-43	Zero classical fermion witness does not erase the graded quantum carrier	W5.8	PASS
AC5-44	No Fock Hilbert completion or fermionic measure is inferred	W5.8	PASS
AC5-45	Spatial generator is the exact improved proper-redescription derivation	W5.9	PASS
AC5-46	Internal Gauss generator acts on every SM field sector	W5.9	PASS
AC5-47	All six spin(1,3) Lorentz generators act on the coframe and chiral-spinor sectors	W5.9	PASS
AC5-48	Exact off-shell D plus G plus J convention remains intact with a spin(1,3)-valued connection correction	W5.9	PASS
AC5-49	All linear constraints are smeared in declared classes including boosts and rotations	W5.9	PASS
AC5-50	Gravitational kinetic term uses the exact inherited supermetric convention	W5.10	PASS
AC5-51	Metric-dependent ordering and density coordinates are printed	W5.10	PASS
AC5-52	Gravitational potential and cosmological convention are exact	W5.10	PASS
AC5-53	Scalar-clock kinetic and potential terms are complete	W5.10	PASS
AC5-54	Gauge and Higgs Hamiltonian terms are complete on the SM branch	W5.10	PASS
AC5-55	Fermionic Hamiltonian is represented by smooth strong transposes continuous degree-wise dGamma lifts and separately typed finite ordering components	W5.10	PASS
AC5-56	Derived torsion-free spin connection is not replaced by an independent variable	W5.10	PASS
AC5-57	Every Hamiltonian candidate retains epsilon and ordering indices	W5.10	PASS
AC5-58	Primitive totality is proved on a degree-wise complete Bastiani-smooth exterior-section algebra before unguarded domain closure	W5.11	PASS
AC5-59	Every smeared constraint including the complete branch-B fermionic and section-valued Hessian actions is a total endomorphism of the common domain and has a transpose on the algebraic dual	W5.11	PASS
AC5-60	Algebraic action is not promoted to Hilbert-domain density or invariance	W5.11	PASS
AC5-61	Operator compositions adjoints and closures remain explicit	W5.11	PASS
AC5-62	WDW-6 rows Formal core joint-solution functor is precisely defined from total transposed actions	W5.12	PASS
AC5-63	Formal SM joint-solution functor includes D G all six J and the completed-carrier Hamiltonian constraint	W5.12	PASS
AC5-64	Formal solution spaces are allowed to be zero	W5.12	PASS
AC5-65	Formal solution functor is not labeled H_phys	W5.12	PASS
AC5-66	Candidate equivalence uses a bicontinuous filtered graded ambient map plus explicit invertible base groupoid representation-algebra and primal-dual one-particle transports	W5.12	PASS
AC5-67	Equal-looking formal equations are insufficient for candidate equivalence	W5.12	PASS

ID	Condition	Proof gate	Adjudication
AC5-68	Complete fixed-rung and smooth-core representations regulator pivot topologized exterior-section carrier ambient domain ordering density chart and clock coordinates index the family	W5.13	PASS
AC5-69	No omitted coordinate is declared skin without the topology-bearing two-level matching package and total bijection of complete constraint labels	W5.13	PASS
AC5-70	Core and SM branches remain separately typed within one exported package	W5.13	PASS
AC5-71	All sixteen QReg5 clauses have theorem witnesses in the composable matching-package result	W5.14	PASS
AC5-72	All twenty-two exact nonclaims are printed	W5.14	PASS
AC5-73	WDW-6 continuation ledger is complete and target indexed	W5.14	PASS
AC5-74	No anomaly inner-product observable or regulator-removal burden is reported closed	W5.14	PASS
AC5-75	AASC necessity and conventional positive proof labor remain separated	W5.14	PASS
AC5-76	Conclusion stops at the candidate-family handoff to WDW-6	W5.14	PASS

Appendix F. Adversarial map

F.1 Hostile-attack adjudication

ID	Attack	Required closure	Proof gate	Adjudication
ADV5-01	The paper merely renames the ADM-to-WDW substitution	Construct configuration carriers differentials regulators domains and typed actions	W5.2-W5.11	CLOSED
ADV5-02	AASC is being used as a primitive quantization rule	Separate kernel necessity from every positive quantum-candidate construction	W5.1; W5.14	CLOSED
ADV5-03	A successful candidate is used to authorize the kernel	Enforce no-backflow dependency and exact import order	W5.1	CLOSED
ADV5-04	The finite-rank spacetime connection theorem is silently generalized to superspace	Prove the type-jump exclusion and independent field-space calculus	W5.6	CLOSED
ADV5-05	Riemannian metrics are treated as a linear space globally	Use the open positive cone chart and state its topology	W5.2	CLOSED
ADV5-06	A globally smooth superspace is assumed despite stabilizers	Use groupoid equivariance and local regular slice only	W5.3	CLOSED
ADV5-07	Improper or boundary transformations are erased	Retain target guard and prohibit quotient outside proper compact branch	W5.3	CLOSED
ADV5-08	The probe algebra fails to separate configuration directions	Prove probe separation sector by sector	W5.5	CLOSED
ADV5-09	Compact support on an infinite-dimensional chart is invoked falsely	Use cylinder/local support discipline without local-compactness assumption	W5.5	CLOSED
ADV5-10	The functional derivative is only a symbol	Construct first and second Frechet differentials with domain and codomain	W5.4-W5.5	CLOSED
ADV5-11	Continuous duals are identified with pointwise fields without proof	Print density pairing and any Riesz or distributional identification	W5.4	CLOSED
ADV5-12	The coincident second derivative remains undefined	Type L2-Hr-H-r-H-r-L2; prove trace class; forbid an H-r-to-Hr multiplier	W5.7	CLOSED
ADV5-13	The regulator hides a preferred background	Register h_star profile and transport as candidate coordinates	W5.7; W5.13	CLOSED
ADV5-14	The DeWitt supermetric normalization is guessed rather than imported	Trace all factors signs and densities to the accepted canonical convention	W5.4; W5.10	CLOSED
ADV5-15	Ordering ambiguity is concealed in notation	Print the ordering ledger and retain mathfrak O on every Hamiltonian	W5.10	CLOSED
ADV5-16	The scalar clock is treated as external time	Keep it as constrained matter and preserve its failure boundary	W5.10; W5.14	CLOSED
ADV5-17	The local clock is promoted to global preferred or quantum time	State exact nonclaim and WDW-7 boundary	W5.14	CLOSED
ADV5-18	The classical zero-fermion witness is used to delete fermions	Construct the full graded CAR test module and fermionic constraint terms	W5.8; W5.10	CLOSED
ADV5-19	Fermionic second-class reduction is ignored	Seed the CAR from the accepted Dirac-reduced bracket	W5.8	CLOSED
ADV5-20	A Fock representation or fermionic measure is assumed without construction	Keep completion determinant and measure in the WDW-6 ledger	W5.8; W5.14	CLOSED
ADV5-21	The WDW-4B off-shell bracket repair regresses	Keep D plus G plus J with the spin(1,3)-valued correction; pure D only weakly	W5.9	CLOSED
ADV5-22	Internal Gauss or Lorentz response omits a field sector	Audit every carrier sector and all six Lorentz directions	W5.9	CLOSED
ADV5-23	Unmeared operator-valued distributions are reported as operators	Use only declared smearings and label kernels as distributions	W5.9-W5.10	CLOSED
ADV5-24	The Hamiltonian omits a matter or clock contribution	Audit both branches against the full classical constraint ledger	W5.10	CLOSED

ID	Attack	Required closure	Proof gate	Adjudication
ADV5-25	The derived torsion-free spin connection is quantized as independent data	Keep omega_LC derived and register independent connection as target update	W5.2; W5.10	CLOSED
ADV5-26	The common domain is defined by assuming the operators already exist	Construct the completed exterior-section carrier and prove strong-transpose dGamma ordering and target-valued Hessian primitives total before unguarded closure	W5.11	CLOSED
ADV5-27	Algebraic invariant-domain action and transpose are promoted to a dense Hilbert operator	Print proof-class status and WDW-6 domain burdens	W5.11; W5.14	CLOSED
ADV5-28	Closure under individual actions is confused with closure under compositions	Evaluate all finite words in total completed-carrier primitives; leave topological products and quantum commutators to WDW-6	W5.11	CLOSED
ADV5-29	The formal joint solution set is called a physical Hilbert space	Use Sol5 in Phi dual and reserve H_phys	W5.12	CLOSED
ADV5-30	Nonemptiness of the joint kernel is assumed	Allow Sol5 to be zero and export nontriviality as WDW-6 burden	W5.12	CLOSED
ADV5-31	Constraint algebra closure is inferred from the classical bracket	Separate classical structure functions from quantum commutators	W5.12; W5.14	CLOSED
ADV5-32	Local Standard Model anomaly cancellation is promoted to quantum-gravity anomaly freedom	Retain the full target-indexed anomaly continuation ledger	W5.14	CLOSED
ADV5-33	Two candidates are called equivalent because their formal equations look alike	Require a bicontinuous filtered graded complete matching with explicit mu_s mu_infinity one-particle base groupoid and bijective complete constraint-label maps plus componentwise identity inverse composition	W5.12-W5.13	CLOSED
ADV5-34	Regulator or ordering changes are dismissed as skin before testing solutions	Classify them as candidate tensors until faithful equivalence is proved	W5.13	CLOSED
ADV5-35	Core and Standard Model branches are conflated	Preserve distinct carriers actions solution functors and indices	W5.2; W5.13	CLOSED
ADV5-36	Core and Standard Model branches are disconnected from the integrated handoff	Trace both through the exact 4A-to-4B interface and lift	W5.1; W5.13	CLOSED
ADV5-37	Connection loop BRST reduced or group-averaged formalisms are declared equivalent by analogy	Register them only as comparison branches requiring intertwiners	W5.13	CLOSED
ADV5-38	A kinematical inner product is silently treated as physical	Keep all inner products candidate or open until WDW-6	W5.14	CLOSED
ADV5-39	Topology bundle boundary or singular sectors are claimed classified	Print exact target guard and nonclaims	W5.1-W5.2; W5.14	CLOSED
ADV5-40	Numerical couplings or Planck-scale data are attributed to AASC	Retain inherited parameters as fixed target coordinates	W5.1; W5.14	CLOSED
ADV5-41	Ledger status substitutes for proof	Trace status to the pivot six-Lorentz strong-transpose dGamma complete-target trace primitive-totality compact-open topology two-level seed-action proofs and the theorem-level composable matching tuple	W5.14	CLOSED
ADV5-42	WDW-5 steals WDW-6 closure burdens	Stop the conclusion at the typed regulator-indexed candidate family	W5.14	CLOSED